

Structural Principle of Entanglement (PSE)

*Foundations of Emergence:
physical quantities as images of a fundamental pregeometric structure*

Conceptual Framework

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Abstract

The Structural Principle of Entanglement (PSE) proposes an emergentist framework in which no physical quantity is fundamental. Spacetime, dynamics, fields, statistical distributions, and all physical observables emerge from a real prespatial structure S , devoid of coordinates, metric, causality, and physical properties. The fundamental reorganization $\mathfrak{D}$ of the prespatial structure S generates the discrete fundamental structural distance $D_S^{(f)}$ and the selective binary correlation C_f , regulated by the ontological threshold Δ . The same fundamental reorganization $\mathfrak{D}$, continuing its action on the fundamental structure S , produces the continuous emergent structural distance $D_S^{(e)}$ and the emergent correlation C_e , which constitute a continuous but non-observable pregeometric domain. It is the action of $\mathfrak{D}$ that generates the increase in complexity of the emergent structure: when such complexity exceeds a threshold, the emergent configurations become representable as physical quantities. The map Φ , monotonic, non-invertible, and compressive, precisely formalizes that level of complexity at which the emergent structure becomes physically representable:

$$\Phi: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow \mathcal{D}_{\text{physical}}.$$

Distinct classes of maps Φ_α formalize different physical domains (geometric, dynamical, statistical, field-theoretic, or discrete), all derived from the same structural information. The informational rigidity of Φ prevents physical quantities from amplifying the emergent correlation, ensuring internal coherence between local and non-local phenomena. Physical constants emerge as compressed images of structural constraints: the threshold Δ and discreteness imply a minimal correlation scale ($\hbar$), the structural coherence limit c becomes the relativistic limit, while the compressivity of Φ determines a response coefficient that in the physical domain takes the form of the gravitational constant G .

Emergence follows a unidirectional and non-invertible ontological chain:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F\lambda} D_S^{(e)} \xrightarrow{G\lambda} C_e \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}.$$

Physical observables are effective representations of the emergent correlation: none can reconstruct or influence the fundamental structure. Local and global phenomenology depends on structural variations $\Delta D_S^{(e)}$: local variations generate domains compatible with experimental physics, while global variations produce effective large-scale structures, including the geometric and dynamical representations used in cosmology.

The PSE provides a unified description of the emergence of physical quantities, grounded in the compressivity of the map Φ and consistent with the main contemporary lines of research on pregeometry, information, and the emergence of spacetime.

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Part I - General Introduction

1 Why the name “Structural Principle of Entanglement”

The term *Structural Principle of Entanglement* (PSE) derives from the ontological nature of entanglement within the PSE: it is not a physical property, nor a quantum phenomenon, but a structural consequence of the emergent correlation that forms along the ontological chain of the fundamental domain. Entanglement is not interpreted as an interaction, a dynamics, or a physical process, but rather as a structural constraint imposed by the emergent correlation C_e and the emergent structural distance $D_S^{(e)}$.

The principle is “structural” because it concerns exclusively the pregeometric structure and its regularizations, independently of any physical representation; it is “of entanglement” because the emergent correlation it describes coincides, in the physical domain, with what is observed as entanglement.

The name therefore formalizes the fact that entanglement is a compressed image of a non-physical ontological correlation, and that its origin lies in the emergent structure, not in the dynamics of the physical domain.

2. Reality in the PSE

In the PSE, S is the primitive ontological domain: a real, prespatial level devoid of any physical structure. It is composed of pure relational entities, elements that possess no intrinsic properties and no geometric or dynamical characteristics. A pure relational entity exists only insofar as it is part of a network of relations with other entities of S ; it has no internal identity, no state, no form.

What can change are not the entities themselves, but the relations that connect them. The fundamental reorganization $\mathfrak{D}$ acts exclusively on the relations, generating different configurations of S . From these configurations emerges the fundamental structural distance $D_S^{(f)}$, the first organizational level that prepares the formation of the emergent structure.

2.1 Reality as an Emergent Structure

The fundamental reality described by the PSE does not coincide with the physical domain, nor with the set of measurable quantities. It is constituted by the emergent structure, that is, by the complex configuration of correlations that derive from the configuration of the prespatial structure S . This structure, represented by $D_S^{(e)}$ and C_e , is real in the strongest sense: it exists autonomously, independently of any observer and of any physical interpretation. Reality is not defined by physical quantities, but by the coherence and stability of the emergent relations that constitute nature itself.

2.2 Emergent Complexity, Discrete World, Macroscopic Objects, and Cosmological Structures

As complexity increases, the emergent structure generates the discrete world, namely the level at which emergent correlations acquire stable, distinguishable, and persistent configurations. The discrete world is not yet a physical domain, but constitutes the first structured form of reality: a set of coherent configurations that manifest as pre-physical discrete units. The discrete units of the discrete world are not entities nor objects: they are coherent configurations of the emergent correlation. They possess no mass, charge, spin, position, or any physical property; they are not localized and do not exist as individual elements. The discrete world is a structural level, not a collection of objects: its

units are patterns of coherence, not physical components. At this level, elementary configurations appear that physics will interpret as “particles” only after the quantum compression performed by Φ_{quant} ; within the discrete world they are not physical entities, but simple structural units.

Macroscopic objects are not aggregates of physical entities, nor collections of particles or fields. In the PSE, they are complex and stable configurations of the discrete world: emergent patterns that maintain coherence through the structural index t , understood as a simple ordering of fundamental configurations. The elementary configurations of the discrete world are not physical particles: they are pre-physical structural units, lacking mass, spin, charge, or any quantum property. Only after the quantum compression performed by Φ_{quant} can these configurations be represented as particles within the physical domain.

Macroscopic objects are more complex extensions of the elementary configurations of the discrete world: stable forms of the emergent correlation that maintain rigidity and coherence on larger scales. Their existence does not depend on physical quantities, but on the structural stability of the emergent correlation. In this sense, macroscopic objects are real but not physical: they belong to emergent reality, not to its physical representation.

The same gradient of complexity that generates discrete units and macroscopic objects continues on increasingly larger scales, producing global configurations of the emergent correlation. When such configurations achieve large-scale coherence, the map Φ represents them as geometric and dynamical structures: curvature, effective metrics, and collective behaviors that physics interprets as cosmological phenomena. The global properties of the Universe (expansion, rigidity, distribution of structures) are therefore compressed images of structural variations in the emergent domain, not fundamental dynamics. Cosmology thus represents the macroscopic form of the same emergent reality that, at lower complexity, produces elementary configurations and objects.

2.3 Reality Is Observable, but Only Beyond a Complexity Threshold

The emergent structure, in its elementary forms, is non-observable in both physical and experiential senses: it produces neither measurable quantities nor perceivable forms. However, when the fundamental reorganization generates sufficiently high levels of relational complexity, the emergent structure becomes accessible to perception. What we see, touch, and recognize is not physics, but the emergent structure at a level of complexity capable of producing objects, forms, and stable regularities.

The essential distinction is therefore threefold:

- **non-observable physically**, because it lacks observables;
- **non-observable experientially at elementary levels**, because it lacks form;
- **observable experientially at complex levels**, where the emergent structure produces perceivable configurations.

The map Φ formalizes what emergent complexity already makes perceivable.

2.4 Reality as an Autonomous and Pre-Physical Domain

Reality in the PSE is an autonomous ontological domain, which does not depend on physics nor on its categories. It exists prior to physics, independently of any quantitative representation. Physics is not reality, but a language that interprets it. The emergent structure constitutes nature; physics is its translation. This distinction is fundamental for understanding the role of the map Φ and the non-physical nature of the fundamental levels of the PSE.

2.5 Why Reality Must Be Separated from Physics

Traditional physics tends to identify reality with what is measurable. The PSE overturns this perspective: reality is what spontaneously emerges from the configurations of the structure S , whereas physics is an effective representation obtained through Φ . Without this distinction, the physical reader would interpret the emergent structure as “pre-physical” or “proto-spatial,” completely misunderstanding the nature of the theory. The conceptual separation between reality and physics is therefore necessary to preserve the ontological coherence of the PSE.

This distinction makes it possible to understand why the PSE is not a physical theory, but a theory of reality: physics does not coincide with reality, but constitutes a compressed, monotonic, and non-invertible translation of it. Having clarified that the reality of the PSE is constituted by the emergent structure and that physics represents only an effective image of it, we can now introduce the conceptual framework that formalizes this reality in its fundamental elements.

3. Distinction Between Reality and Physics

3.1 Reality and Physics as Distinct Ontological Domains

In the PSE, reality and physics belong to different ontological domains. Reality is constituted by the emergent structure, that is, by the complex configuration of correlations C_e and emergent distances $D_S^{(e)}$ that derive from the structure S . This structure is autonomous, spontaneous, and independent of any interpretation. Physics, by contrast, is not an ontological domain: it is a representational language that describes reality through measurable quantities. The distinction between these two levels is essential for understanding the nature of the PSE.

3.2 Physics as an Effective Representation of Reality

Physics does not coincide with reality, but constitutes an effective representation of it. Physical quantities (distances, times, fields, metrics) do not exist in the emergent domain: they appear only after the application of the map Φ , which interprets the emergent structure as a physical domain. Physics is therefore a system of description, not a fundamental level of being. It translates emergent reality into measurable quantities, but does not generate or modify it.

3.3 The Map Φ as a Bridge Between Reality and Physics

The map Φ is the conceptual element that connects emergent reality to the physical domain. It is not real, not a dynamics, and not a physical process: it is an interpretative transformation that makes physically readable what is already real. Φ does not add information, does not introduce new quantities, and does not alter the emergent structure. Its function is exclusively representational: it allows reality to be expressed in physical terms without modifying its ontological nature.

3.4 Why Reality Is Deeper Than Physics

Emergent reality is deeper than physics because it exists independently of any representation. Physics depends on the map Φ , whereas reality does not. The emergent structure is autonomous, spontaneous, and complete: it does not require observables, metrics, or fields in order to exist. Physics, instead, requires a translation of reality into measurable quantities. In this sense, physics is a derived level, whereas reality is the fundamental level.

3.5 Conceptual Consequences for the Interpretation of the PSE

The distinction between reality and physics prevents the PSE from being interpreted as a traditional physical theory. The emergent structure is not a “pre-space,” nor a “proto-field,” nor a hidden physical

domain: it is reality itself. The discrete world is not a set of physical entities, but a complex level of emergent reality. Macroscopic objects are not aggregates of particles, but stable configurations of emergent correlation. Physics describes these objects only after the projection Φ , but does not create them.

3.6 Necessity of the Conceptual Separation

Without a clear separation between reality and physics, the physical reader would interpret the emergent structure through physical categories, misunderstanding the nature of the PSE. The distinction is therefore necessary to preserve the coherence of the theory: reality emerges spontaneously from the fundamental reorganization $\mathcal{D}$ on S , whereas physics is an effective representation obtained through the projection map Φ .

This separation makes it possible to understand why the PSE is not a physical theory, but a theory of reality: physics does not coincide with reality, but constitutes a compressed, monotonic, and non-invertible translation of it. Having clarified that the reality of the PSE is constituted by the emergent structure and that physics represents only an effective image of it, we can now introduce the conceptual framework that formalizes this reality in its fundamental elements.

Part II - Conceptual Framework

Contemporary physics is described by extremely effective but conceptually incompatible theories, because they assume geometric, dynamical, or statistical structures as primitive. General relativity presupposes a continuous and metric spacetime, whereas quantum mechanics describes discrete, non-local phenomena lacking intrinsic geometry. This tension arises from the idea that observable physical structures are fundamental.

The Structural Principle of Entanglement (PSE) rejects this assumption: no physical quantity is fundamental. All observables emerge from a deeper prespatial structure, S , devoid of coordinates, metric, causality, dynamics, or physical properties. The elements of S are defined only by their structural relations, described by the fundamental structural distance $D_S^{(f)}$. This choice is motivated by the epistemic need to describe non-local correlations without assuming underlying geometry or dynamics.

Through the fundamental structural reorganization $\mathfrak{D}$ on S , the discrete fundamental structural distance $D_S^{(f)}$ generates the selective binary correlation C_f , regulated by the threshold Δ . The collective aggregation of fundamental correlations produces a continuous emergent structural distance $D_S^{(e)}$, belonging to a non-observable but already regular domain. From $D_S^{(e)}$ derives the continuous emergent correlation $C_e(i, j)$, which represents the pregeometric form of coherence and constitutes the level from which all physical quantities will arise. C_e does not live in spacetime: its nonlocality is a structural, not dynamical, property, and allows non-local physical correlations to be understood without introducing physical transport or superluminal mechanisms.

The fundamental level is not physicalist: it contains no physical quantities, possesses no observable dynamics, and cannot be described using geometric, dynamical, or statistical concepts. Physics emerges only as an effective representation of the successive structural levels.

Physical quantities emerge through a monotonic map

$$\Phi: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow \mathcal{D}_{\text{physical}}$$

which projects the emergent structure into a physical domain. The same structure $(X, D_S^{(e)}, C_e)_{\text{reg}}$ generates distinct physical domains through different classes of maps Φ_α . Each class produces an effective representation (geometric, dynamical, field-theoretic, statistical, or discrete), ensuring that all physical structures emerge from the same structural information. Multi-emergence guarantees coherence among the different physical domains and prevents conflicts between geometry, dynamics, fields, and quantization. The map Φ is compressive: it reduces structural information to a physical representation, which is why no emergent quantity can reconstruct the prespatial structure.

The ontological hierarchy is:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F\lambda} D_S^{(e)} \xrightarrow{G\lambda} C_e \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}.$$

and is regulated by the fundamental reorganization $\mathfrak{D}$, which incorporates the structural index t , the structural coherence limit c , and the principle $E + C$, and by the threshold Δ , which selects the fundamental correlations that contribute to C_f .

From these constraints physical constants also emerge: discreteness and the threshold Δ produce a minimal scale that appears as the Planck constant; the structural limit c becomes the relativistic limit;

the compressivity of Φ generates a response coefficient that in the physical domain takes the form of the gravitational constant G . None of these constants is fundamental: all are compressed images of pregeometric constraints.

The hierarchy is non-retroactive: each level emerges from the previous one without influencing it, and no physical quantity can reconstruct the prespatial structure from which it derives.

The PSE shows that observable physics is not a set of primitive axioms, but the compressed manifestation of a real prespatial structure. The fundamental reorganization $\mathfrak{D}$ acts exclusively on S : it orders and reorganizes the configurations of the prespatial structure, from which all physical quantities emerge as transformations of the same structural information. The PSE does not modify existing theories: it clarifies their ontological origin, showing that geometry, dynamics, causality, and quantum correlations are emergent representations of the same prespatial structure, obtained through the compressive map Φ .

Observability is not a fundamental property of reality, but a characteristic that emerges when the emergent structure reaches a level of complexity sufficient to produce stable and perceivable forms. Emergent reality exists independently of the map: it is real, structured, and, at elementary levels, non-observable.

The map Φ formalizes the portion of the emergent structure that, thanks to its complexity, can be represented as measurable physical quantities in the physical domain.

Technical Note

The tension between the geometric structures of general relativity and the non-local correlations of quantum mechanics is widely recognized in contemporary literature. General relativity assumes a continuous and metric spacetime (Misner, Thorne & Wheeler 1973), whereas quantum mechanics describes discrete, non-local phenomena lacking intrinsic geometry (Bell 1964; Brunner et al. 2014). This incompatibility has motivated approaches that question the fundamentality of observable physical structures. In causal sets (Bombelli et al. 1987; Sorkin 2003), geometric structures emerge from discrete relations. In non-commutative geometry (Connes 1994; Connes & Marcolli 2008), the fundamental structure is algebraic, and physical quantities emerge from spectral properties. In informational models (Van Raamsdonk 2010; Swingle 2012), physical structures are interpreted as images of more primitive correlations. In this context, the PSE's idea that all observables emerge from a prespatial structure S is fully consistent with contemporary research directions.

1. Basic Ontological Definitions

In the PSE, the term *real* denotes an ontologically existing entity, independently of whether it can be observed or measured in the physical domain. *Real* does not coincide with *observable*: an entity may be real without being physical, local, or measurable. The fundamental prespatial structure S is real in this ontological sense.

An entity is *observable* when it can be measured, detected, or inferred through physical interactions in the emergent domain, that is, in the domain formalized by the projection:

$$\Phi: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow \mathcal{D}_{\text{physical}}.$$

Observability is an emergent property: it appears only when a structural quantity is transformed into a physical quantity through the map Φ .

An entity is *non-observable* when it possesses no direct physical interaction with the emergent domain, while still contributing to its structure through ontological correlations. The following are

non-observable: S , the fundamental structural distance $D_S^{(f)}$, and the emergent structural distance $D_S^{(e)}$.

The fundamental domain is a real and non-observable pregeometric structure, containing ontological correlations devoid of physical interpretation. It contains no physical distances, no metrics, no operators, no states, no locality, and no physical interactions, and for this reason it cannot be observable.

The emergent domain is the transformation of the fundamental correlations into a continuous domain X on which the emergent structural distance $D_S^{(e)}$ and the emergent correlation C_e are defined. It is still non-observable, yet already structured: it contains continuous relational information while possessing no physical quantities. X represents the emergent threshold of complexity that allows the pre-geometric structure to acquire the continuity and stability required for the projection Φ . It is the first level at which reality can be defined, while still not being physical.

The physical domain is the level at which observables and geometric, dynamical, statistical, field-theoretic, or discrete structures appear, depending on the class of map Φ . It is defined by the projection Φ , which transforms the pregeometric structure into measurable quantities.

Technical Note

The distinction between real and observable is consistent with ontological and informational approaches in contemporary physics. In non-commutative geometry (Connes 1994; Connes & Marcolli 2008), there exist real entities that are not directly observable, whose structure determines emergent physical properties. In causal sets (Bombelli et al. 1987; Sorkin 2003; Dowker 2020), the discrete structure is real but non-observable: only the emergent physical quantities are measurable. In informational models (Chiribella, D'Ariano & Perinotti 2011; Van Raamsdonk 2010; Swingle 2012), ontological correlations do not live in spacetime and are not observable, yet they determine emergent physical properties. In this context, the PSE distinction between $S(\text{real})$, $D_S^{(f)}$ and $D_S^{(e)}$ (non-observable), and $\Phi(X, D_S^{(e)}, C_e)_{\text{reg}}$ (observable) is fully consistent with contemporary literature on pregeometry and the emergence of physical structures.

1.1 Ontological Chain of Emergence

The ontological structure of the PSE is organized into a chain of transformations that leads from the non-observable ontology to the emergent physical quantities. The chain is not geometric, dynamical, or statistical: it is a sequence of structural levels, each of which contains less information than the previous one.

The fundamental prespatial structure S contains all possible ontological correlations. It is not physical, not local, not measurable, but it is real in the ontological sense defined earlier. From S emerges the fundamental structural distance $D_S^{(f)}$, which defines the fundamental domain, a non-observable pregeometric structure devoid of any physical quantity. Within it, the fundamental correlations C_f are organized, still lacking any physical interpretation.

From these correlations arises the emergent structural distance $D_S^{(e)}$, which defines the emergent domain. In it, the fundamental correlations are transformed into the emergent correlations C_e , which anticipate physical structure and contain continuous relational information, while remaining non-observable.

Physical quantities emerge through a monotonic map:

$$\Phi: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow \mathcal{D}_{\text{physical}},$$

which projects the pregeometric structure into the physical domain. The map Φ can generate geometric, dynamical, statistical, field-theoretic, or discrete structures, depending on the projection class. In this passage, observables, measurable interactions, and experimental structures appear.

The complete ontological chain is therefore:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F\lambda} D_S^{(e)} \xrightarrow{G\lambda} C_e \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}.$$

Observability is not a fundamental property, but an emergent one that arises when the emergent structure reaches a sufficient level of complexity. It does not belong to the ontological structure, but to its physical representation. The map Φ formalizes, in physical form, the portion of the emergent structure that has already acquired sufficient complexity and stability to be represented as a measurable quantity. The prespatial structure remains real but not measurable.

The hierarchy is non-retroactive: each level emerges from the previous one without influencing it, and no physical quantity can modify or reconstruct the prespatial structure from which it derives, because it contains less information and represents a compressed projection of the emergent domain.

Technical Note

The ontological chain of the PSE is consistent with emergentist models in which physics arises from successive transformations of non-geometric structures. In causal sets (Bombelli et al. 1987; Sorkin 2003; Dowker 2020), geometric structures emerge from discrete relations through coarse-graining processes. In tensor network models (Swingle 2012; Qi 2013), physical structures are obtained from emergent correlations through monotonic maps, indicating that physics can be interpreted as a compressed image of a deeper informational structure. In emergent gravity models (Jacobson 1995; Padmanabhan 2010), physical dynamics is interpreted as an effective representation of microscopic non-geometric properties. In this context, the chain of the PSE stands as a coherent, non-invertible, and non-retroactive emergent structure, fully compatible with contemporary research lines on pregeometry and the emergence of physical structures.

1.2 Why the Fundamental Domain Cannot Be Observable

The fundamental domain is pregeometric: it contains no physical distances, no metrics, no operators, no states, no locality, and no physical interactions. By definition, it cannot be measured, detected, or observed, because observability requires the existence of emergent physical quantities, which appear only after the projection through the map:

$$\Phi: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow \mathcal{D}_{\text{physical}}.$$

The fundamental domain possesses no physical structure: it contains no geometric, dynamical, statistical, or field-theoretic quantities. Consequently, it cannot be observable, and the physical domain emerges only subsequently. Observability is an emergent property that arises when the emergent structure reaches a sufficient level of complexity; it is not an original attribute of the ontological structure. The map Φ merely formalizes, in physical form, the portion of the emergent structure that has already acquired sufficient complexity and stability to be represented as a measurable quantity.

The fundamental domain is real but non-observable: it contributes to the structure of reality through ontological correlations, but it cannot be measured or inferred by any physical interaction. Only the

projection Φ makes measurement possible, transforming the pregeometric structure into physical quantities.

Technical Note

The idea that the fundamental level is non-observable is consistent with models in which physical structures emerge only at the effective level. In causal sets (Bombelli et al. 1987; Sorkin 2003; Dowker 2020), the microscopic level lacks physical quantities and cannot be observed: only emergent quantities are measurable. In non-commutative geometry (Connes 1994; Connes & Marcolli 2008), the fundamental structure is real but non-observable, and physical quantities emerge from spectral properties. In spin network and spin foam models (Markopoulou 2000; Oriti 2016), fundamental states possess no coordinates, no causality, and no physical quantities, and cannot be measured. In this context, the fundamental domain of the PSE is necessarily non-observable: observability requires the projection Φ , which generates the physical domain from the pregeometric structure.

2. Structural Principle of Entanglement

2.1 Short Version

Fundamental reality is a prespatial relational structure S , devoid of coordinates, metric, causality, and physical properties. The fundamental reorganization $\mathfrak{D}$ acts exclusively on S and generates the discrete fundamental structural distance $D_S^{(f)}$, non-observable and non-physical. No physical quantity exists at the fundamental level: geometry, dynamics, causality, fields, and statistical distributions emerge as effective representations of the emergent structural distance $D_S^{(e)}$, which is likewise non-observable. Observability arises only when the emergent structure reaches a sufficient level of complexity and is formalized in the physical domain through the compressive map Φ .

Technical Note

The interpretation of physical structures as emergences of non-geometric correlations is consistent with several contemporary approaches to pregeometry. In holographic models (Ryu & Takayanagi 2006; Van Raamsdonk 2010), geometry is reconstructed from quantum correlations. In causal sets (Bombelli et al. 1987; Sorkin 2003; Dowker 2020), physical structures emerge from discrete relations. In tensor network models (Swingle 2012; Qi 2013), continuous structures are images of informational networks. In this context, entanglement as the physical manifestation of structural proximity is fully consistent with the idea that non-local correlations derive from pregeometric structures.

2.2 Extended Version

We assume the existence of a fundamental prespatial structure S , composed of elements lacking coordinates, position, metric, causality, dynamics, or physical properties. The fundamental reorganization $\mathfrak{D}$ acts exclusively on S and generates the discrete fundamental structural distance $D_S^{(f)}(i, j)$, which:

- is not geometric,
- does not belong to a physical domain,
- does not presuppose causality,
- reflects configurations ordered by a discrete non-physical structural index.

From $D_S^{(f)}$ derives the fundamental correlation $C_f(i, j)$, binary and selective, regulated by the coherence threshold Δ . The collective aggregation of fundamental correlations produces the emergent

structural distance $D_S^{(e)}(i, j)$, continuous and non-observable, from which derives the emergent correlation $C_e(i, j)$, a monotonic function of $D_S^{(e)}$.

Physical quantities emerge through a monotonic and compressive map:

$$\Phi: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow \mathcal{D}_{\text{physical}},$$

which formalizes the pregeometric structure in the physical domain. The map Φ can generate geometric, dynamical, statistical, field-theoretic, discrete, and quantum structures. Physical entanglement is the manifestation of the emergent correlation C_e , not of physical distance or geometry.

Technical Note

The idea that quantum correlation derives from a pregeometric structure is consistent with fundamental results in quantum information theory. Bell's theorem (1964) and the CHSH inequalities (Clauser, Horne, Shimony & Holt 1969) show that no local theory can reproduce quantum correlations. Tsirelson's bound (1980) establishes the maximum value of quantum correlations compatible with the theory. In holographic models (Maldacena & Susskind 2013), entanglement is interpreted as a pregeometric structure determining connections between physical regions. In this context, the emergent correlation C_e of the PSE stands as the structural basis of physical entanglement.

2.3 Ontological Hierarchy of Levels

- **Fundamental level:** prespatial structure S , fundamental distance $D_S^{(f)}$, fundamental correlation C_f .
- **Emergent structural level:** dominio emergente continuo X , emergent distance $D_S^{(e)}$, emergent correlation C_e .
- **Physical level:** domain $\mathcal{D}_{\text{physical}}$ generated by Φ , containing geometric, dynamical, statistical, field-theoretic, discrete and quantum structures

The hierarchy is unidirectional: each level emerges from the previous one without feedback. Each level contains less information than the previous one. The physical domain is an effective representation of the emergent domain, which in turn is a representation of the fundamental domain. It is not possible to reconstruct the fundamental structure from any emergent physical quantity.

Technical Note

The ontological hierarchy of the PSE is consistent with emergentist models in which each level contains less information than the previous one. In causal sets (Dowker 2020), physical structures contain less information than the original discrete structure. In holographic models (Ryu & Takayanagi 2006), geometry is a compression of quantum correlations. In emergent gravity models (Jacobson 1995; Padmanabhan 2010), physics is interpreted as a macroscopic phenomenon derived from microscopic non-geometric structures. In this context, the PSE hierarchy stands as a coherent, non-invertible, and non-retroactive emergent structure.

2.4 Fundamental Axioms

Axiom 1 — Structural Fundamentality

There exists a real prespatial structure S , which constitutes the primary ontological level. The elements of S possess no geometric, dynamical, statistical, or physical properties; they are defined exclusively by the structural relations that emerge from the fundamental reorganization $\mathfrak{D}$.

Axiom 2 — Fundamental Reorganization $\mathfrak{D}$

The prespatial structure S is subject to a fundamental reorganization $\mathfrak{D}$, which orders and reorganizes structural configurations. The reorganization $\mathfrak{D}$ incorporates three structural components:

- the discrete structural index t , non-physical and non-metric, which orders configurations without defining any dynamics;
- the structural coherence limit c , which constrains admissible variations between adjacent configurations without introducing velocity or physical propagation;
- the $E + C$ principle, which regulates the stability of fundamental configurations through the minimization of contradictions and the reduction of tensions.

The reorganization $\mathfrak{D}$ is a local property of the structure S and generates the fundamental structural distance $D_S^{(f)}$.

Axiom 3 — Fundamental Distance and Correlation

On S is defined the fundamental structural distance $D_S^{(f)}(i, j)$, pregeometric and discrete. From it derives the fundamental correlation $C_f(i, j)$, binary and selective, regulated by the threshold Δ . The collective aggregation of the fundamental correlations, regulated by Δ , forms the basis for the emergence of the emergent structural distance $D_S^{(e)}$, generated by the same fundamental reorganization $\mathfrak{D}$.

Axiom 4 — Emergent Distance and Correlation

From the fundamental level emerges an emergent structural distance $D_S^{(e)}$, continuous, regular, and compressive with respect to $D_S^{(f)}$. From the emergent distance derives the emergent correlation $C_e(i, j)$, defined through a monotonic function:

$$C_e(i, j) = F(D_S^{(e)}(i, j)),$$

where F is neither universal nor invertible. Both $D_S^{(e)}$ and C_e belong to the pregeometric domain and constitute the minimal emergent structures required for the definition of the emergent domain X

Axiom 5 — Emergence of the Physical Domain

Physical quantities emerge through a monotone and compressive map:

$$\Phi: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow \mathcal{D}_{\text{fisico}}.$$

The physical domain does not exist at the fundamental level: it is an effective representation of the emergent structure. Φ does not *create* observability; it formalizes it, translating the regularized emergent structure into physical quantities.

The raw emergent structure

$$(X, D_S^{(e)}, C_e)$$

is first regularized through the Structural Regularization Operator:

$$R: (X, D_S^{(e)}, C_e) \rightarrow (X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}}),$$

and subsequently filtered by the complexity thresholds:

$$\Theta_k: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}}),$$

which ensure that the emergent structure satisfies the minimum complexity requirements necessary for physical projection.

Only after the combined application of R and the thresholds Θ_k can the map Φ operate in a **coherent**, continuous, and monotone manner.

Axiom 6 — Emergent Physical Structures

Discrete, geometric, dynamical, statistical, field-like, and quantum structures do not exist at the fundamental level. They appear in the emergent physical domain as compressed representations of the emergent structural distance $D_S^{(e)}$ and the emergent correlation C_e .

These structures are defined on the regularized emergent structure:

$$\left(X, D_S^{(e)}, C_e\right)_{\text{reg}},$$

where the regularization is performed by the Structural Regularization Operator

$$R: \left(X, D_S^{(e)}, C_e\right) \rightarrow \left(X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}}\right),$$

and subsequently filtered by the complexity thresholds

$$\Theta_k: \left(X, D_S^{(e)}, C_e\right)_{\text{reg}} \rightarrow \left(X, D_S^{(e)}, C_e\right)_{\text{reg}}.$$

The projection map

$$\Phi: \left(X, D_S^{(e)}, C_e\right)_{\text{reg}} \rightarrow \mathcal{D}_{\text{phys}}$$

does not introduce new structures; it **translates** the regularized pregeometric emergent structure into effective physical quantities. Physical structures are therefore **representational**, not fundamental: they emerge solely from the compressed image of the pregeometric emergent relations.

Axiom 7 — Emergent Limits

The structural limits of the fundamental domain — including the limit on admissible variations c , the minimal granularity $\hbar$, the minimal rigidity (mass), and the total rigidity (energy) — are not physical in nature. When the emergent structure is represented in the physical domain through the projection Φ , these limits appear as constraints on observable physical quantities, taking the form of the characteristic constants and bounds of relativistic and quantum phenomenology. Φ does not introduce these limits; it **translates** them into effective physical constraints.

Technical Note

The structural axioms of the PSE are consistent with pregeometric and informational approaches to quantum gravity. The axiom of structural fundamentality is compatible with discrete models (Bombelli et al. 1987; Markopoulou 2000). The axiom of the structural index is consistent with timeless approaches (Barbour 1999) and with the “thermal time hypothesis” (Connes & Rovelli 1994). The axiom of fundamental correlation aligns with logical-relational (Barwise & Seligman 1997) and informational models (Chiribella, D’Ariano & Perinotti 2011). The axiom of emergent correlation is consistent with models of geometry from correlations (Van Raamsdonk 2010; Swingle 2012). The axioms on emergent physical structures and emergent limits are consistent with models in which physical quantities derive from discrete structures (Hossenfelder 2018; Trugenberger 2017–2024).

3. Ontological Consequences

3.1 Fundamental and Emergent Structural Distance

Structural distance is an internal property of the prespatial structure S . It is not a geometric distance, does not presuppose space, time, causality, or metric structure, and does not correspond to any physical notion of separation. It is an ontological relation that measures **structural proximity** between elements or configurations of the structure. It appears in two distinct forms: a fundamental one and an emergent one.

Fundamental Structural Distance $D_S^{(f)}$

$D_S^{(f)}$ is the distance defined between the fundamental elements of the prespatial structure.

Characteristics:

- **Discrete:** it takes discrete, non-continuous values.
- **Pregeometric:** it is not defined within any spatial framework; it is an internal relation of S .
- **Non-physical:** it does not correspond to any observable distance.
- **Selective:** it determines which pairs of elements may contribute to the fundamental correlation C_f .
- **Relational:** it measures structural coherence between elements, not spatial positions.
- **Local:** the fundamental reorganization $\mathfrak{D}$ acts only on relations that are close in the sense of $D_S^{(f)}$.

In summary: $D_S^{(f)}$ is an ontological distance between fundamental elements, entirely devoid of geometric meaning.

Emergent Structural Distance $D_S^{(e)}$

$D_S^{(e)}$ is the distance defined between emergent configurations of the structure, i.e., coherent patterns composed of many fundamental elements.

Characteristics:

- **Aggregated:** it results from many fundamental relations.
- **Continuous:** it appears continuous as a collective effect of aggregation.
- **Coherent:** it measures structural proximity between stable emergent configurations.
- **Pregeometric:** it is not yet a physical distance; it belongs to the emergent but still non-observable level.
- **Derived:** it is constructed from $D_S^{(f)}$, the fundamental correlation C_f , and the selectivity threshold Δ .

In summary: $D_S^{(e)}$ is a distance between emergent patterns, not a distance in physical space.

Relation Between the Two Distances

The emergent distance is a transformation of the fundamental distance:

$$D_S^{(e)}(P, Q) = F(D_S^{(f)}(i, j))_{i \in P, j \in Q},$$

where F is an aggregation function consistent with the selectivity imposed by Δ . The threshold Δ determines which fundamental correlations are admissible and therefore which contributions enter the construction of $D_S^{(e)}$.

Associated Correlations

- **Fundamental correlation** C_f — binary, discrete, selective; defined on fundamental elements.
- **Emergent correlation** C_e — continuous, aggregated; defined on emergent configurations.

Both correlations are pregeometric and non-observable

Technical Note

The distinction between fundamental and emergent distance is consistent with models in which physical structures derive from discrete relations. In causal sets (Bombelli et al. 1987; Dowker 2020), fundamental distance is discrete, while continuous distance emerges from aggregation processes. In spin network models (Markopoulou 2000; Oriti 2016), fundamental distance is defined by discrete relations, while continuous distance is a collective effect. In informational models (Van Raamsdonk 2010; Maldacena & Susskind 2013), emergent distance is a function of correlations, analogous to the role of C_e in the PSE.

3.2 Threshold Δ and Correlation Selectivity

The threshold Δ is an ontological constraint internal to the prespatial structure S . It is not a numerical value, not a physical parameter, and cannot be tuned or assigned: Δ is a property of the structure, not a constant. Its role is to determine which fundamental relations are sufficiently coherent to be admissible.

The fundamental structural distance $D_S^{(f)}$ is **discrete**: it does not take continuous values and has no geometric interpretation. Consequently, Δ does **not** compare metric magnitudes and does **not** impose “small” or “large” thresholds. Δ acts directly on the **structural coherence** of fundamental relations: a relation between two fundamental elements is admissible only if the discrete configuration of $D_S^{(f)}$ satisfies the constraint imposed by Δ . In this case the fundamental correlation is present ($C_f = 1$); otherwise it is absent ($C_f = 0$).

Δ does not evaluate magnitudes or intensities: it selects only discrete patterns of structural proximity. A fundamental relation is admissible only if its discrete configuration of $D_S^{(f)}$. Only admissible fundamental correlations contribute to the construction of the emergent distance $D_S^{(e)}$ and the emergent correlation C_e . The discreteness of the fundamental domain does not arise from any physical process, but from this structural constraint: Δ determines the very form of the fundamental level and its ability to generate a coherent emergent structure.

The selectivity induced by Δ produces **rigidity**, **saturation**, and **regularity** in physical quantities, preventing uncontrolled divergences and ensuring that the physical domain is bounded and stable. This mechanism is analogous to structural filtering and informational coarse-graining, in which only significant correlations survive structural compression.

Δ guarantees that the emergent structure is regular and non-pathological. The selectivity of correlations is a fundamental property of the ontological hierarchy, not a physical effect.

Technical Note

Correlation selectivity is consistent with models in which emergent structure derives from filtered or thresholded correlations. In holographic models (Van Raamsdonk 2010; Maldacena & Susskind 2013), only sufficiently strong correlations contribute to the reconstruction of emergent geometry. In tensor networks (Swingle 2012; Qi 2013), emergent distance is determined by the strength of correlations, which act as a structural threshold. In informational models (Chiribella & Scandolo 2015), distance is defined as a monotonic function of selected correlations. In coarse-graining emergence models (Bény 2013; Ojima 2004), selectivity is essential for obtaining regular, divergence-free structures. In this context, the PSE's Δ stands as an ontological threshold determining which fundamental correlations participate in constructing the emergent structure.

3.3 Fundamental and Emergent Correlation

The fundamental correlation C_f is binary, selective, and pregeometric. It is determined solely by the fundamental structural distance $D_S^{(f)}$ and regulated by the coherence threshold Δ , which specifies which fundamental relations are admissible. C_f has no autonomous dynamics: every variation of C_f arises exclusively from local reorganizations of the fundamental structural distance. It is not a physical quantity, not observable, and does not constitute a hidden variable.

The emergent correlation C_e is defined by:

$$C_e(i, j) = F \left(D_S^{(e)}(i, j) \right),$$

where F is a monotonic function that translates the emergent distance into a degree of structural correlation.

Characteristics of C_e :

- **continuous**, because it aggregates many fundamental relations;
- **regular**, because it inherits the coherence of $D_S^{(e)}$;
- **pregeometric and non-observable**, because it belongs to the emergent non-physical domain;
- **structural**, because it does not yet represent any physical quantity.

C_e does not represent any attenuated or preliminary form of physical correlation: it is a purely structural correlation, devoid of physical meaning until the projection Φ . Only Φ makes part of the emergent correlation physically describable, without altering its ontological nature.

Ontological role:

- it links the fundamental distance $D_S^{(f)}$ to the physical domain;
- it determines physical quantities through the monotonic map Φ ;
- it provides the structural basis for physical entanglement, which emerges as a compressed image of non-observable emergent correlations.

Technical Note

The interpretation of C_e as a monotonic function of emergent distance is consistent with models in which physical structures derive from correlations. In holographic models (Van Raamsdonk 2010; Maldacena & Susskind 2013), physical structures are reconstructed from monotonic correlations. In tensor networks (Swingle 2012; Qi 2013), correlation is the basis of emergent distance. In

informational models (Chiribella & Scandolo 2015), distance is defined as a monotonic function of correlations. In this context, the PSE's C_e stands as a conceptual bridge between prespatial structure and the physical domain.

3.4 Relation Between Levels

The structural emergent distance $D_S^{(e)}$ derives from the structural fundamental distance $D_S^{(f)}$ through the fundamental correlation C_f and the selectivity threshold Δ . The emergent correlation C_e is a monotonic function of $D_S^{(e)}$. Physical quantities are compressed images of the emergent structure, obtained through the projection Φ . Φ is not an ontological entity nor a process: it is a representational relation that does not modify the emergent structure, but provides a compressed physical translation of it.

Each level is completely determined by the previous one and exerts no feedback on the fundamental levels: the hierarchy is strictly unidirectional.

The non-invertibility of the hierarchy is not a technical or computational limitation, but an ontological constraint: each level contains less information than the one before it. The physical domain does not possess, by its very nature, the amount of information required to reconstruct the emergent or fundamental structure.

Technical Note

The unidirectional hierarchy is consistent with emergentist models in which physical structures cannot reconstruct the microscopic structure. In causal sets (Sorkin 2003; Dowker 2020), emergent structures are non-invertible. In holographic models (Ryu & Takayanagi 2006; Van Raamsdonk 2010), maps reconstructing physical structures from correlations are monotonic and non-invertible. In emergent gravity models (Jacobson 1995; Padmanabhan 2010), physical structures do not retroact on the fundamental level. In this context, the PSE hierarchy stands as a coherent and non-retroactive ontological structure.

3.5 Emergence of Physical Representations

In the PSE, no physical quantity is fundamental. Observable magnitudes emerge when the regularized emergent structure

$$\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}$$

reaches a sufficient level of complexity to generate stable and regular configurations.

Regularization is performed by the Structural Regularization Operator:

$$R\left(X, D_S^{(e)}, C_e\right) \rightarrow \left(X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}}\right),$$

and subsequently filtered by the complexity thresholds:

$$\Theta_k: \left(X, D_S^{(e)}, C_e\right)_{\text{reg}} \rightarrow \left(X, D_S^{(e)}, C_e\right)_{\text{reg}}.$$

Physics does not describe the fundamental domain: it represents only its regularities through projection maps.

The projection

$$\Phi: \left(X, D_S^{(e)}, C_e\right)_{\text{reg}} \rightarrow \mathcal{D}_{\text{phys}}$$

formalizes this passage: it does not add information, does not introduce new quantities, and does not modify the emergent structure. It compresses pregeometric correlation into measurable physical forms, translating structural complexity into distances, metrics, fields, dynamics, and distributions.

Physical quantities exist ontologically as properties of the emergent pattern, but they become describable only through the representation provided by Φ .

There are six fundamental classes of emergence maps — geometric, dynamical, statistical, field-like, discrete, and quantum — which generate all effective physical domains. Maps that produce energy or symmetries do not constitute new classes: they are specializations or compositions of the six fundamental ones.

In this framework, the notion of “substance” is not primary. Mass, objects, matter, and persistence do not belong to the fundamental domain: they are effective representations of the structural coherence of the emergent pattern. Physical reality is therefore a compressed image of the pregeometric structure, and every observable property reflects limits, regularities, and internal tensions of the emergent correlation.

The ontological function of the maps Φ is twofold: they make the emergent structure observable and determine the physical form it assumes. Physics does not create reality: it organizes it into measurable domains.

Technical Note

The interpretation of physical quantities as monotonic images of the emergent structure is consistent with models in which physical structures derive from correlations and compression maps. In holographic models (Van Raamsdonk 2010; Maldacena & Susskind 2013), geometry emerges from monotonic correlations. In tensor networks (Swingle 2012; Qi 2013), distance and metric emerge from hierarchical correlation structures. In informational reconstructions (Chiribella & Scandolo 2015), physical quantities are monotonic functions of fundamental correlations. In quantum coarse-graining models (Bény 2013; Petz 1986), physical variables emerge as non-invertible compressions. In this context, the PSE maps Φ_α stand as ontological transformations generating distinct physical domains from the same emergent structure.

3.6 Ontological Role of the Projection Map Φ

The projection map Φ plays an essential role in the PSE: it formalizes, in physical form, the portion of the emergent structure

$$\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}$$

that has reached a sufficient level of complexity.

Regularization is performed by the Structural Regularization Operator:

$$R: \left(X, D_S^{(e)}, C_e\right) \rightarrow \left(X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}}\right),$$

and subsequently filtered by the complexity thresholds:

$$\Theta_k: \left(X, D_S^{(e)}, C_e\right)_{\text{reg}} \rightarrow \left(X, D_S^{(e)}, C_e\right)_{\text{reg}}.$$

Only after these operations does the emergent structure possess the regularity required to be projectable into the physical domain.

Physical quantities are compressed images of the emergent distance and correlation: they are not fundamental properties, but effective representations of the pregeometric structure.

Φ is neither a physical law nor a dynamical operator: it is a monotone, nonlinear, and non-invertible structural transformation that reflects the irreversible loss of fundamental information. The projection introduces no autonomous physicality; it represents the emergent structure in physical form without adding ontological content.

Φ performs no computational operation and does not *calculate* physical quantities: they are not the result of a process, but the compressed representation of the emergent structure. The projection does not modify what it represents; it merely provides a coherent physical form of it.

The non-invertibility of Φ defines the ontological boundary between the emergent structure and the physical domain: what appears as “physical” is a compressed representation of the pregeometric structure, not a reconstruction of it. No physical quantity is fundamental; all are monotone and compressed images of the emergent correlation.

Technical Note

The ontological role of Φ is consistent with models in which physical structures emerge from monotonic transformations and information loss. In informational reconstructions (Chiribella & Scandolo 2015), physical quantities derive from monotonic maps that compress more fundamental structures. In quantum coarse-graining models (Bény 2013; Petz 1986), information loss is essential for the emergence of effective variables. In relational interpretations (Rovelli 1996), physical observables are non-fundamental representations of deeper structures. In non-metric ontological models (Henson 2015), observable physical structure is a projection of a pregeometric structure. In this context, the PSE's Φ stands as an ontological map separating emergent structure and physical domain.

3.7 Fundamental Hierarchy of the PSE

The complete ontological hierarchy is:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F\lambda} D_S^{(e)} \xrightarrow{G\lambda} C_e \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}.$$

The fundamental reorganization $\mathfrak{D}$ acts on the relational configurations of S , generating the fundamental structural distance $D_S^{(f)}$. From $D_S^{(f)}$ emerges the fundamental correlation C_f . Emergent regularization of the relations produces the emergent structural distance $D_S^{(e)}$, from which the emergent correlation C_e arises.

The raw emergent structure

$$(X, D_S^{(e)}, C_e)$$

is then regularized through the Structural Regularization Operator:

$$R: (X, D_S^{(e)}, C_e) \rightarrow (X, D_{S_{\text{reg}}}^{(e)}, C_{e_{\text{reg}}}),$$

and filtered by the complexity thresholds:

$$\Theta_k: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (X, D_S^{(e)}, C_e)_{\text{reg}}.$$

Emergence is real: physical quantities do not pre-exist, but appear when the regularized emergent structure

$$\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}$$

reaches a level of complexity sufficient to generate stable, regular, and coherent configurations.

Technical Note

The ontological hierarchy of the PSE is consistent with emergentist models in which each level contains less information than the previous one. In causal sets (Dowker 2020), physical structures contain less information than the original discrete structure. In holographic models (Ryu & Takayanagi 2006), physical structures are compressed representations of correlations. In emergent gravity models (Jacobson 1995; Padmanabhan 2010), physics is a macroscopic effect of microscopic non-geometric structures. In this context, the PSE hierarchy stands as a coherent, non-invertible emergent structure.

4. Structure of the Domains

The theory distinguishes three ontological domains, each characterized by different quantities, properties, and conceptual meaning. Each domain emerges from the previous one through non-invertible and non-retroactive transformations. This articulation is essential to avoid overlap between levels and to clarify which quantities belong to the prespatial structure and which to emergent physics.

4.1 Fundamental Domain

The fundamental domain is constituted by the prespatial structure S , the fundamental structural distance $D_S^{(f)}$, and the fundamental correlation C_f . Its main characteristics are:

- absence of geometry;
- absence of physical causality;
- absence of continuity;
- non-observability;
- description through a discrete, non-physical structural index;
- presence of the structural coherence limit c .

In this domain, only structural relations exist between elements lacking coordinates or physical properties. The fundamental reorganization $\mathfrak{D}$ acts locally on the relational configurations of S , generating the fundamental structural distance $D_S^{(f)}$. From $D_S^{(f)}$, through the coherence threshold Δ , derives the fundamental correlation C_f , binary and selective. No physical quantity belongs to this level: the fundamental domain is real but completely pregeometric and non-observable.

Technical Note

The distinction of a fundamental domain devoid of physical quantities is consistent with several contemporary approaches to pregeometry. In causal sets (Bombelli et al. 1987; Sorkin 2003; Dowker 2020), the microscopic level consists of discrete elements without coordinates, and physical structures emerge only at the effective level. In non-commutative geometry (Connes 1994; Connes & Marcolli 2008), the fundamental structure is algebraic and possesses no physical quantities: such properties emerge from spectral operators. In dynamic spin network models (Markopoulou 2000; Oriti 2016), the fundamental state is discrete, non-metric, and non-observable, and dynamics acts on elementary relations. In this context, the PSE's fundamental domain stands as a discrete prespatial

structure, devoid of physical quantities and governed by a local fundamental reorganization acting on $D_S^{(f)}$.

4.2 Non-Observable Emergent Domain

From the fundamental domain there emerges a continuous and regular structural level, consisting of the emergent domain X on which the following quantities are defined:

- the emergent structural distance $D_S^{(e)}$;
- the continuous emergent correlation $C_e(i, j)$.

This domain is pregeometric: it is not yet physical, but it possesses sufficient continuity and regularity to allow the definition of physical quantities through the projection Φ . Its characteristics are:

- $D_S^{(e)}$ is an aggregated function of many fundamental relations;
- C_e is a monotonically decreasing function of $D_S^{(e)}$;
- no quantity is observable;
- no physical structure exists yet;
- emergent correlation is coherent but not geometric.

This domain is the conceptual bridge between the prespatial structure and emergent physics.

Technical Note

The existence of a continuous but non-observable emergent domain is consistent with models in which continuity arises as a collective effect of discrete structures. In causal sets (Bombelli et al. 1987; Dowker 2020), continuity emerges from coarse-graining processes but is not yet physical. In spin-network condensate models (Gielen & Oriti 2016; Chirco & Oriti 2017), a continuous pregeometric level exists that precedes emergent physics. In informational models (Van Raamsdonk 2010; Maldacena & Susskind 2013), continuity derives from the structure of correlations, while physical quantities emerge only after an effective projection. In this context, the PSE's emergent domain stands as a continuous, coherent, but non-physical pregeometric level.

4.3 Observable Physical Domain

The observable physical domain arises from the increase in complexity of the non-observable emergent domain. When the emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}$ surpasses a complexity threshold sufficient to produce stable, coherent, and regular configurations, it becomes possible to represent it through the monotonic projection map Φ , which does not generate the emergence of physical quantities but formalizes them, translating the pregeometric emergent structure into effective physical magnitudes:

$$\Phi: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow \mathcal{D}_{\text{phys}}.$$

Within the physical domain appear:

- observable physical quantities;
- geometric, dynamical, statistical, field-like, or discrete structures;

- the quantum structure $(\mathcal{H}, A, \hbar)$ as a compressed image of the emergent correlational constraints;
- physical causality as a compressed image of emergent coherence;
- experimental phenomenology.

Characteristics of the physical domain:

- it is the only observable domain;
- it possesses physical structures;
- physical causality is an effective representation of emergent correlation formalized by Φ ;
- the relativistic limit is the compressed image of the structural limit c ;
- physics is an effective representation of $D_S^{(e)}$ and C_e , not a reconstruction of the emergent structure.

Technical note

Interpreting the physical domain as a compressed representation of a pregeometric structure is consistent with several emergentist approaches. In holographic models (Ryu & Takayanagi 2006; Van Raamsdonk 2010), physical structures are reconstructed from correlations through monotonic, non-invertible maps. In emergent-gravity models (Jacobson 1995; Padmanabhan 2010), physical structures are effective representations of microscopic non-geometric properties. In tensor-network frameworks (Swingle 2012; Qi 2013), physical structures emerge as continuous limits of discrete networks. Within this landscape, the physical domain of the PSE stands as the observable level made accessible by emergent complexity and formalized by Φ .

4.4 Relations Between Domains

The complete hierarchy is:

$$\left(\mathcal{S} \xrightarrow{\mathcal{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \right) \xrightarrow{F\lambda} \left(D_S^{(e)} \xrightarrow{G\lambda} C_e \right) \xrightarrow{\Phi} \mathcal{D}_{\text{fisico}}.$$

Properties of the hierarchy:

- **unidirectional:** no level retroacts on the previous one;
- **non-invertible:** no physical quantity can reconstruct the prespatial structure;
- **non-local:** emergent correlation is not constrained by any physical structure;
- **non-physical in the first two levels:** only the domain generated by Φ is observable;
- **coherent:** each level inherits regularity from the previous one.

This structure avoids physicalist interpretations of the fundamental level and clarifies that physics is an emergent representation, not a set of primitive axioms.

Technical Note

The unidirectional and non-invertible hierarchy between levels is consistent with models in which physical structures cannot reconstruct the microscopic structure. In causal sets (Sorkin 2003; Dowker 2020), emergent structures do not allow reconstruction of the original discrete structure. In holographic models (Ryu & Takayanagi 2006; Van Raamsdonk 2010), maps reconstructing physical structures from correlations are monotonic and non-invertible. In emergent gravity models (Jacobson

1995; Padmanabhan 2010), physical structures do not retroact on the microscopic level. In informational models (Chiribella, D'Ariano & Perinotti 2011), emergent correlation does not live in any physical domain. In this context, the PSE hierarchy stands as a coherent and non-retroactive ontological structure.

Technical Note on the Quantum Structure

The quantum structure $(\mathcal{H}, A, \hbar)$ does not constitute an additional ontological domain: it is an effective representation obtained through the compression map Φ_{quant} , which translates the correlational constraints of the emergent domain into states, operators, and minimal thresholds. Quantization is not a fundamental property of reality, but an effect of the saturation of the emergent correlation and of the compressive nature of the projection.

This interpretation is consistent with several contemporary approaches to emergent quantum mechanics. In informational models (Chiribella, D'Ariano & Perinotti 2011), quantum states are effective representations of more primitive correlational structures. In holographic models (Maldacena 1997; Van Raamsdonk 2010), the quantum structure emerges from monotone maps that compress non-local correlations. In tensor network models (Swingle 2012; Qi 2013), the state space is an effective representation of discrete correlational constraints. Within this framework, the quantum structure of the PSE is positioned as a compressed image of the emergent correlation, not as a separate ontological level.

5. Fundamental Reorganization $\mathfrak{D}$

The fundamental reorganization $\mathfrak{D}$ is the structural process that rewrites the prespatial structure S along the discrete structural index t . It acts exclusively on the fundamental structure S , reorganizing the relations among elements that possess no physical properties. “The ‘action’ of $\mathfrak{D}$ does not imply any dynamics, evolution, or temporal process: $\mathfrak{D}$ does not modify S over time, but defines which configurations of the structure are admissible, coherent, and compatible with the ontological constraints. The reorganization is therefore a structural constraint, not a physical change.

The reorganization is local within the structure generated by $D_S^{(f)}$: admissible variations between adjacent configurations are limited by the structural coherence parameter c , which introduces no notion of physical velocity or propagation. $\mathfrak{D}$ follows the structural principle $E + C$, which minimizes contradictions and reduces tensions among nearby relations.

$\mathfrak{D}$ is not a physical dynamics, not geometric, and not defined in any physical domain: it is a pregeometric structural process that precedes every form of physics and constitutes the primary ontological level from which all subsequent structures emerge.

5.1 Structural Index t

The structural index t is a discrete parameter that orders the fundamental configurations of the prespatial structure:

$$S_t \text{ with } t \in \mathbb{N}$$

It is not a physical time, it has no metric, it is not continuous, and it does not define causality. It serves exclusively to label the configurations of the prespatial structure. The index t does not represent any form of evolution: it is a simple ordering, devoid of temporal or dynamical meaning. Referring to “configurations along t ” only indicates the adjacency relation between admissible configurations, not a process in time. The fundamental reorganization $\mathfrak{D}$ is defined as a local property with respect to $D_S^{(f)}$:

every variation of the structure involves only structural relations that are close in the sense of $D_S^{(f)}$. This reorganization is a property of the structure, not a process in time.

Technical Note

The interpretation of the structural index t as an internal property of the pregeometric structure is consistent with several contemporary approaches to pregeometry and rewriting dynamics. In such models, the fundamental transformation is described as a rewriting of states, devoid of any physical temporal variable. In computational models, the transformation is represented as a discrete sequence of local updates; in the thermal time hypothesis, observable time is interpreted as an effective parameter derived from internal transformations of the state. In this context, the PSE's structural index introduces no ontological dynamics: it orders the configurations of the prespatial structure without defining succession, duration, or causality. Physical time arises only after the projection Φ , as a compressed image of the emergent structure.

5.2 Structural Coherence Limit c

The fundamental reorganization $\mathfrak{D}$ is subject to a structural coherence limit c , which regulates the amplitude of admissible modifications of the fundamental structural distance $D_S^{(f)}$. The limit c arises from the discrete nature of the prespatial structure S : since S is composed of elementary units and rewriting rules with finite support, every transformation between adjacent configurations can involve only a finite number of elementary units. It follows that there exists a maximum extent for the admissible reorganizations in a single ontological step.

Since S possesses no physical time, this reorganization is not a temporal evolution: it is a discrete structural property. The parameter c is therefore the fundamental constant that constrains the maximum amplitude of transformations between configurations of the prespatial structure. Its universality in the physical domain derives from the fact that c is an intrinsic property of the discrete prespatial structure, not of the physical domain.

Characteristics of c :

- it is not a physical velocity;
- it is not defined in any physical domain;
- it does not imply causality;
- it is a constraint on the reorganization of the fundamental structure.

The relativistic limit observed in the physical domain is the image of c through the projection Φ .

Technical Note

The interpretation of the fundamental limit c as a structural constraint is consistent with several contemporary approaches to discrete pregeometry. In discrete models, state transformations are limited by the granularity of the substrate; in causal sets, causality and the limiting speed emerge from discreteness, not from fundamental metric properties. Lieb–Robinson bounds show that there exists a maximum speed of correlational influence in discrete systems, independently of the presence of a physical metric. In dynamic spin-network models, analogous limits emerge from the combination of discreteness and locality.

In this context, the PSE's structural limit c does not represent a physical velocity nor a fundamental dynamics: it is an internal constraint on the discrete reorganization of the prespatial structure, limiting the amplitude of admissible variations between adjacent configurations in the structural index. Its

image in the physical domain, through the projection Φ , coincides with the observable relativistic constant.

5.3 Principle $E + C$

The fundamental reorganization $\mathfrak{D}$ is governed by a structural minimization principle:

$$E + C = \text{reduction of contradictions} + \text{reduction of tensions.}$$

- E represents the internal coherence of the structure: the reorganization tends to eliminate incompatible configurations.
- C represents local compatibility: the reorganization tends to reduce tensions among nearby relations.

This principle is not physical: it does not minimize energy, does not define equations of motion, and does not imply temporal continuity. It is a structural criterion that guides the reorganization of $D_S^{(f)}$.

Technical Note

The $E + C$ principle finds analogies in models of self-organization, complex networks, and logical-relational frameworks. In such contexts, stability emerges from the reduction of internal incompatibilities and from local compatibility, without relying on fundamental physical quantities. The PSE's $E + C$ principle stands as a general structural rule governing the reorganization of $D_S^{(f)}$.

5.4 Reorganization of $D_S^{(f)}$

The fundamental reorganization $\mathfrak{D}$ produces local transformations of the structure, rewriting the fundamental structural distance between adjacent configurations of the discrete structural index. Each transformation:

- modifies the fundamental correlation C_f ;
- alters local coherence;
- contributes to the emergence of $D_S^{(e)}$;
- does not generate any direct physical phenomena.

The fundamental reorganization is discrete, non-continuous, and non-physical. It is the process from which emergent regularity arises.

Technical Note

The local reorganization of the prespatial structure presents analogies with models of emergent geometry from discrete structures, coarse-graining processes, and quantum network transformation models. In such frameworks, continuity and coherence emerge from discrete local transformations. In this context, the PSE's fundamental reorganization is not an ontological dynamics: it is a local structural property of the fundamental distance $D_S^{(f)}$, ensuring the coherence of configurations ordered by the discrete structural index. This property produces the continuity of the emergent distance $D_S^{(e)}$ without introducing succession, causality, or evolution in the fundamental domain.

5.5 Emergence of $D_S^{(e)}$

The emergent structural distance $D_S^{(e)}$ arises from the collective regularization of the fundamental reorganizations.

Characteristics:

- it is continuous;
- it is coherent;
- it is non-observable;
- it is non-physical;
- it is the basis of the emergent correlation C_e .

The fundamental reorganization does not act directly on $D_S^{(e)}$: it is an emergent effect of the reorganization of $D_S^{(f)}$.

Technical Note

The idea that a continuous distance may emerge from a discrete structure is supported by causal sets, emergent-geometry models based on correlations, and spin-network condensate approaches. In this context, the PSE's $D_S^{(e)}$ stands as a collective regularization effect, non-observable and non-physical.

5.6 Emergence of the Emergent Correlation C_e

The emergent correlation is a monotonically decreasing function:

$$C_e(i, j) = F\left(D_S^{(e)}(i, j)\right).$$

The fundamental reorganization $\mathfrak{D}$ does not act on C_e : variations of C_e derive exclusively from variations of $D_S^{(e)}$.

Technical Note

Models of emergent geometry based on correlations, holographic approaches, and relational informational models show that physical structures can be derived from monotonic functions of correlation. In this context, the PSE's C_e stands as a pregeometric structure derived from $D_S^{(e)}$.

5.7 Emergent Operators F_λ and G_λ

The emergence of continuous structural quantities in the emergent domain requires the introduction of two families of regularization operators, F_λ and G_λ , which act on the discrete fundamental quantities $D_S^{(f)}$ and C_f . These operators do not represent any physical or geometric dynamics: they are purely structural transformations that produce progressively more regular and continuous versions of the fundamental quantities, in accordance with the approximation scale λ .

The operator F_λ acts on the fundamental structural distance:

$$F_\lambda: D_S^{(f)} \rightarrow D_\lambda,$$

where D_λ is a regularized distance defined on the approximated domain S_λ .

Similarly, the operator G_λ acts on the fundamental correlation:

$$G_\lambda: C_f \rightarrow C_\lambda,$$

where C_λ is a regularized correlation defined on the same domain.

The families $\{F_\lambda\}_{\lambda \in \Lambda}$ and $\{G_\lambda\}_{\lambda \in \Lambda}$ are monotonic, compressive, and stable: for $\lambda_2 > \lambda_1$, the functions $F_{\lambda_2}(D_S^{(f)})$ and $G_{\lambda_2}(C_f)$ are more regular than the corresponding approximations at smaller scale. Compressivity ensures that each transformation reduces the information contained in the fundamental quantities, while stability ensures that local variations compatible with the structural limit

$$|D_S^{(f)}(t+1) - D_S^{(f)}(t)| \leq c$$

produce controlled variations in the regularized quantities D_λ and C_λ .

The limits

$$D_S^{(e)} = \lim_{\lambda \rightarrow \infty} F_\lambda(D_S^{(f)}), \quad C_e = \lim_{\lambda \rightarrow \infty} G_\lambda(C_f)$$

define respectively the emergent structural distance and the emergent correlation of the domain X . These quantities are continuous, pregeometric, non-observable, and saturated: they do not represent any physical magnitude, but constitute the mathematical basis for the future projection Φ that will generate the physical domain.

The emergent operators F_λ and G_λ represent the formal mechanism through which the fundamental structure is regularized and made continuous, preparing the emergence of the structural quantities $D_S^{(e)}$ and C_e . When these quantities reach a sufficient degree of continuity, coherence, and regularity, they define the non-observable emergent domain X .

5.8 No Physical Dynamics in the Fundamental Levels

In the fundamental and non-observable emergent levels:

- no physical equations of motion exist;
- no physical causality exists;
- no physical propagation exists;
- no metric exists;
- no energy exists;
- no mass exists;
- no gravity exists.

The fundamental reorganization $\mathfrak{D}$ is purely structural: it operates within the discrete structural index t , with structural coherence limit c , and according to the $E + C$ principle, reorganizing exclusively the relations of the prespatial structure S . No physical quantity, no geometry, and no observable dynamics exist in this domain. Physics arises only when the emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}$ reaches the necessary complexity and is projected into the physical domain through the monotonic map Φ , which formalizes the emergence of observables.

Technical Note

The absence of fundamental physical quantities is consistent with timeless approaches, non-commutative geometry, informational models, and spin-network quantum gravity. In this context, the PSE places physics as an emergent phenomenon, not a primitive one.

6. Emergence of Physical Quantities

In the PSE, the emergence of physical quantities is a spontaneous process: it does not require any physical dynamics in the fundamental domain, nor the introduction of additional magnitudes.

Observables appear when the emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}$ reaches a sufficient level of complexity to generate stable, regular, and coherent configurations.

The projection map Φ formalizes the passage from the complex emergent structure to the physical domain, translating emergent regularities into measurable quantities, metrics, and fields.

In this sense:

- the emergence of physical quantities is spontaneous, determined by the internal complexity of the emergent structure;
- Φ is a representation map, not a causal mechanism;
- observables exist ontologically before the map, but cannot be described physically until Φ organizes them within the domain of measurements.

Physics does not create emergent reality: it interprets it.

The map Φ represents the decisive conceptual passage of the theory: it does not add information, does not introduce new magnitudes, and does not modify the emergent structure, but provides an effective representation of it in the observable domain.

Φ is a monotonic, non-invertible, and compressive transformation:

- it orders physical quantities according to the structural order of $D_S^{(e)}$;
- it reduces pregeometric information to a measurable form;
- it prevents observables from indefinitely amplifying emergent correlation.

Φ is not a geometric map, but a class of transformations that generate different physical domains (geometric, dynamical, statistical, field-theoretic, or discrete), all derived from the same structural information.

The role of Φ is twofold:

1. it formalizes observability,
2. it determines which physical form the emergent structure assumes.

Geometry, dynamics, statistical distributions, fields, and discrete structures are not fundamental properties of reality, but compressed images of emergent correlation. Each physical domain is a partial representation of the pregeometric structure, and no map can reconstruct the fundamental level from which it derives.

This section presents the general properties of the map Φ and the main classes of projections that generate the different physical domains. These maps constitute the formal mechanism through which the pregeometric structure becomes physical, and represent the point of contact between the ontological hierarchy of the PSE and observable phenomenology.

6.0 Regularization of the Emergent Domain X

In the conceptual framework of the PSE, the emergent domain is denoted by

$$X = \Psi(S),$$

where S is the fundamental structure. The emergence map

$$\Psi: F \rightarrow X$$

is defined as the composition of the transformations that generate the emergent structural distance $D_S^{(e)}$ and the emergent correlation C_e :

$$\Psi = G_\lambda \circ F_\lambda \circ \Delta \circ \mathfrak{D},$$

namely:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e.$$

The raw emergent domain is therefore represented by the structure:

$$(X, D_S^{(e)}, C_e).$$

Although $D_S^{(e)}$ and C_e are already continuous, regular, and saturated (Axiom E4), the raw emergent structure is not yet compatible with the physical compression performed by the projection Φ . To ensure compressive continuity, global stability, and structural coherence, we introduce the **Structural Regularization Operator R**:

$$R: (X, D_S^{(e)}, C_e) \rightarrow (X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}}).$$

R does not modify the domain X : the regularization acts exclusively on the emergent structures $D_S^{(e)}$ and C_e . The regularized structure

$$(X, D_S^{(e)}, C_e)_{\text{reg}}$$

is the only ontological level compatible with:

- the emergent metric

$$g_{\mu\nu} = \Gamma\left((X, D_S^{(e)}, C_e)_{\text{reg}}\right),$$

- the emergent fields

$$\mathcal{F}_{\text{phys}} = \Xi\left((X, D_S^{(e)}, C_e)_{\text{reg}}\right),$$

- the physical quantities

$$Q = \Omega(P),$$

- the derived tensors

$$\mathcal{T}_{\text{der}} = \Lambda(P),$$

- the emergent dynamics

$$\mathcal{D}_{\text{phys}} = \Delta(P).$$

The complexity thresholds

$$\Theta_k: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

determine the minimum level of structural complexity required for the physical projection Φ to be definable. Structural regularization and complexity thresholds constitute the ontological bridge connecting the pregeometric emergent structure to the observable physical domain.

6.1 Emergence Maps

In the PSE, physical quantities emerge through a monotonic map:

$$\Phi: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow \mathcal{D}_{\text{physical}}$$

Φ is not unique: there exist different classes of maps, each of which generates a different physical domain. Geometry is only one of the possible effective representations of the emergent structure.

Φ is not an ontological entity nor a physical process: it is a representational relation that formalizes the emergent structure without modifying it, translating it into physical quantities through a monotonic and non-invertible compression.

Φ possesses four universal properties:

1. Monotonicity

$$D_S^{(e)}(i, j) < D_S^{(e)}(k, l) \Rightarrow \Phi(i, j) < \Phi(k, l)$$

2. Non-invertibility

$$\Phi^{-1} \text{ does not exist}$$

3. Informational Compressivity

$$I\left(\Phi\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}\right) < I\left(D_S^{(e)}, C_e\right)$$

4. Informational Rigidity

No physical quantity can indefinitely amplify the emergent information.

6.2 Classes of Emergence Maps

In the PSE there exist exactly six fundamental classes of emergence maps. Each of them projects the pregeometric structure $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$ into a different emergent physical domain. All other maps appearing in the framework (for example those generating energy, symmetries, or other physical quantities) do not constitute new classes, but are **derived maps**, obtained as specializations or compositions of the six fundamental classes.

The six fundamental classes are:

$$\{\Phi_{\text{disc}}, \Phi_{\text{geom}}, \Phi_{\text{dyn}}, \Phi_{\text{stat}}, \Phi_{\text{field}}, \Phi_{\text{quant}}\}$$

They represent all the ways in which the emergent structure can generate effective physical domains, **with the sole exception of Φ_{disc}** . The discrete map is a special class: it does not generate a physical domain, but a **pre-physical representation** (graphs, networks, discrete structures) of the emergent

correlation. Φ_{disc} is therefore a fundamental map, but not a physical one: it provides a structural reading of the pregeometry, from which the other maps can derive effective physical representations.

(1) Discrete Maps

$$\Phi_{\text{disc}}: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow (G, \Gamma, \mathcal{N})$$

The discrete map Φ_{disc} is the only map that does not lead to the physical domain: it produces a discrete structural representation of the emergent level, useful for informational and discrete models, but devoid of physical meaning.

Technical Note: Causal sets, spin networks, informational models, and tensor networks show that graphs and networks can emerge from more primitive correlations. The discrete map of the PSE fits within this line: G , Γ , and $\mathcal{N}$ are compressed images of $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$.

(2) Geometric Maps

$$\Phi_{\text{geom}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow (d, g_{\mu\nu}, \mathcal{C})$$

Technical Note: Geometry as an effective representation of a pregeometric structure is consistent with holographic approaches, tensor networks, causal sets, and emergent gravity. In these frameworks, distance, metric, and causality are not fundamental but emerge from more primitive correlations. The geometric map of the PSE fits within this line: d , $g_{\mu\nu}$, and $\mathcal{C}$ are compressed images of $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$, while the fundamental structure remains devoid of coordinates, metric, and physical causality.

(3) Dynamical Maps

$$\Phi_{\text{dyn}}: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow (\hat{H}, \hat{U}(T), \rho)$$

Technical Note: Operational approaches, axiomatic reconstructions, relational formulations, and timeless models show that dynamics can emerge from deeper informational structures. The dynamical map of the PSE fits within this line: $\hat{H}$, $\hat{U}(T)$, and ρ are not fundamental entities, but effective representations of $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$.

(4) Statistical Maps

$$\Phi_{\text{stat}}: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow (p(x), S, \langle O \rangle)$$

Technical Note: Informational models, entropic gravity, black-hole phenomenology, and holographic approaches show that entropy and distributions can emerge from non-geometric structures. The statistical map of the PSE fits within this line: $p(x)$, S , and $\langle O \rangle$ are compressed images of $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$.

(5) Field Maps

$$\Phi_{\text{fields}}: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow (\phi(x), A_\mu(x), F_{\mu\nu})$$

Technical Note: Group Field Theory, spin networks, spin foams, informational models, and non-commutative geometry show that fields can emerge from discrete or algebraic structures. The field map of the PSE fits within this line: ϕ , A_μ and $F_{\mu\nu}$ are effective representations of $(X, D_S^{(e)}, C_e)_{\text{reg}}$.

(6) Quantum maps

$$\Phi_{\text{quant}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (\mathcal{H}, A, \hbar)$$

Technical note: Informational approaches, operational reconstructions, algebraic formulations and emergent-quantum models show that quantum structure can be interpreted as an effective representation of deeper correlational patterns. In these frameworks, the state space $\mathcal{H}$, the action A and the minimal threshold $\hbar$ are not fundamental entities, but compressed images of $(X, D_S^{(e)}, C_e)_{\text{reg}}$. The quantum map of the PSE fits within this line: $\mathcal{H}$, A and $\hbar$ are effective representations of the emergent structure, while the fundamental domain remains devoid of operators, non-commutativity and primitive quantization.

6.3 Order of Emergence of the Map Classes

Physical reality does not emerge all at once, but through a continuous progression of complexity that begins at the fundamental structure and unfolds into natural thresholds of emergence. The emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}$ represents the first level at which identities, domains, and coherent correlations appear. As complexity increases, these configurations make possible different forms of representation: first **discreteness** (Φ_{disc}), which captures the ontological separation between regions; then **geometry** (Φ_{geom}), which describes continuity and the emergent metric; next **dynamics** (Φ_{dyn}), which formalizes the coherent evolution of configurations; subsequently **statistics** (Φ_{stat}), which represents collective phenomena; and finally **classical fields** (Φ_{field}), which describe interactions over continuous domains. Only when complexity reaches its maximal threshold does **quantum compression** (Φ_{quant}) become possible, introducing states, operators, a minimal action scale, and stable dynamical modes. Quantum structure is not a separate ontological level, but the final effective representation of the same emergent structure that generated the classical maps. Once they appear, quantum phenomena build the entire physical reality—from atoms to cosmic structures—projecting themselves into the spacetime generated by Φ_{geom} , yet without being ontologically contained within it. Reality is therefore unified and progressive, and the maps are not distinct worlds but **representation thresholds** along the growth of emergent complexity.

6.4 Relationship Between the Map Classes

The emergence order of the map classes does not define an ontological hierarchy nor a functional composition among the maps, but reflects the complexity thresholds of the emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}$. Each map Φ_i becomes definable only when the emergent structure reaches the minimal threshold θ_i required by the domain it represents. The maps do not operate on one another: they all act directly on the same emergent triple and provide compressed representations of it in different physical domains.

The mathematical relationship among the maps is therefore a threshold relation,

$$\theta_{\text{disc}} < \theta_{\text{geom}} < \theta_{\text{dyn}} < \theta_{\text{stat}} < \theta_{\text{campo}} < \theta_{\text{quant}}$$

which expresses the natural progression of complexity required for each map to be well-defined. The maps are compatible sections of the total projection Φ : they do not derive from one another, they do not compose, and they do not form ontological levels, but represent different modes of monotonic compression of the same emergent structure.

6.5 The Role of Quantum Theory in the Hierarchy of Maps

The PSE introduces a decisive conceptual shift in the understanding of the levels of reality. Quantum theory does not belong to the fundamental domain, nor does it represent the original structure of the real: it is the most complex section of the map Φ , since its emergence requires all preceding projections. The quantum representation presupposes a geometric structure, an effective dynamics, a coherent statistical framework, a field-theoretic formalization, and a discrete basis; only after these steps can the emergent correlation be compressed into the form of states, operators, and minimal thresholds.

In this sense, quantum theory occupies a high level in the hierarchy of the Φ maps not because it is “deeper,” but because it is the most compressed and information-dense representation of the emergent correlation. Physics interprets the microscopic domain as an access to the fundamental, but in the PSE it is simply the region of the physical domain in which the compression performed by Φ_{quant} is maximal.

Quantum non-locality is not an intrinsic physical property, nor a propagative phenomenon in space: it is the compressed manifestation of the emergent correlation C_e , filtered through the mathematical structure of the quantum map. The quantum effects observable in the microscopic domain — entanglement, non-local correlations, indeterminacy — do not arise from fundamental dynamics, but from the physical representation of a pregeometric correlation.

Quantum theory is therefore a high-complexity emergent level within the hierarchy of the Φ maps, not the foundation of reality.

6.6 Emergent Physical Quantities

A physical quantity is defined as:

$$Q_{\text{physical}} = \Phi \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

No physical quantity is fundamental. Every observable magnitude is an informational compression of the emergent structure.

6.7 Special Cases

(a) Metric

$$g_{\mu\nu} = \Phi_{\text{geom}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

(b) Physical Time

$$T = \Phi_{\text{geom}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \quad \text{or} \quad T = \Phi_{\text{dyn}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

(c) Physical Dynamics

$$\text{dynamics} = \Phi_{\text{dyn}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

(d) Physical Fields

$$\phi(x) = \Phi_{\text{fields}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

(e) Statistical Distributions

$$P = \Phi_{\text{stat}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

(f) Symmetries

$$\mathcal{G} = \Phi_{\text{sym}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

with

$$\Phi_{\text{sym}} \subset \{ \Phi_{\text{geom}}, \Phi_{\text{dyn}}, \Phi_{\text{stat}} \}$$

\

(g) Energetic Structures

$$E = \Phi_{\text{ener}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

with

$$\Phi_{\text{ener}} \subset \Phi_{\text{dyn}}$$

(h) Quantum structures

$$(\mathcal{H}, A, \hbar) = \Phi_{\text{quant}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

6.8 Mass and Energy as Emergent Structural Properties

In the PSE, mass and energy are neither fundamental quantities nor intrinsic physical properties. They emerge as structural characteristics of the pregeometric domain $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$, and are represented as physical quantities only through the projection Φ . Their nature is therefore ontologically pre-physical and depends exclusively on the properties of the emergent structure.

Mass as Emergent Rigidity of the Pattern

Mass is defined as the *emergent rigidity* of the correlation pattern within the pregeometric domain. It measures the amount of structural coherence required to maintain a configuration stable across emergent time. A configuration endowed with high rigidity resists changes in its emergent form, exhibiting a structural persistence that, in the physical domain, is represented as inertia.

Formally, mass is a property of the emergent triplet $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$, and cannot be defined in the fundamental domain nor in the physical domain without the intervention of the projection Φ . Its existence is therefore a direct consequence of the fundamental reorganization $\mathfrak{D}$ and of the stability of emergent patterns.

Energy as Emergent Tension of the Pattern

Energy is defined as the *emergent tension* of the pattern, namely the resistance to structural relaxation imposed by the $E + C$ principle. It measures how strongly an emergent configuration opposes the reduction of its internal complexity. A highly coherent configuration exhibits high emergent tension, which in the physical domain is represented as energy.

Energy is also a property of the non-observable emergent domain and cannot be defined at the fundamental level. Its physical representation is obtained through the map Φ_{ener} , which translates emergent tension into measurable physical quantities.

Structural Equivalence Between Mass and Energy

In the PSE, mass and energy are not distinct quantities: they are two manifestations of the same structural property, the *emergent coherence* of the pattern. Mass measures resistance to changes in the configuration; energy measures resistance to relaxation. Since both arise from the rigidity and tension of the emergent pattern, their equivalence is an ontologically inevitable consequence of the structure, not a postulated physical principle.

The physical relation between mass and energy is therefore the image, through Φ , of a deeper and pregeometric equivalence. The familiar physical relation between mass and energy is not fundamental, but represents the geometric compression of a structural property of the emergent domain.

Role of the Projection Φ

The projection Φ does not create mass and energy; it provides a physical representation consistent with the emergent geometry. In the physical domain, mass and energy appear as measurable quantities, but their origin is entirely structural. The distinction between mass and energy is therefore an artifact of the physical representation, whereas in the emergent domain they constitute complementary aspects of the same ontological property.

6.9 Other Emergent Properties of the Structure

Beyond mass and energy, the emergent domain $(X, D_S^{(e)}, C_e)_{\text{reg}}$ gives rise to additional properties that, in the physical domain, are represented as fundamental quantities or structures. These are not physical entities at the ontological level: they are characteristics of the form, coherence, and symmetries of the emergent pattern, which the projection Φ translates into measurable physical quantities.

Emergent Spin

Spin is not an intrinsic property of physical objects, but an emergent symmetry of the correlation pattern. Its discrete representation in the physical domain results from the compression performed by the projection maps, which translate coherent configurations into quantized values.

Emergent Charge

Charge is a structural asymmetry of the emergent pattern. It does not exist in the fundamental domain and is not a primary physical quantity: it is a property of the emergent correlation C_e , which Φ represents as a discrete quantity associated with physical fields.

Physical Fields as Compressions

Fields are not ontological entities. They are images of coherent variations in $D_S^{(e)}$ and C_e , compressed by Φ_{field} into continuous and localized structures. Their dynamics are an effective representation of the structural reorganization of the emergent domain.

Emergent Geometry

Geometry is not assumed at the ontological level. It is a compression of the emergent structural distance $D_S^{(e)}$ through Φ_{geom} . The metric properties of the physical domain reflect limits and regularities of the emergent structure, not fundamental characteristics of space.

Emergent Dynamics

Physical dynamics are an effective representation of variations in the emergent pattern. They do not exist in the fundamental domain and cannot be assumed as primary properties. They result from the compression of structural transformations into local and temporal physical laws.

Emergent Symmetries

Physical symmetries are images of structural regularities in the emergent domain. They are not fundamental properties of physics, but consequences of the coherence of the pattern and the stability of correlations. Φ translates these regularities into geometric and gauge symmetries.

Emergent Physical Constants

Physical constants (c , $\hbar$, G) are not fundamental parameters. They are structural limits of the emergent domain, represented in the physical domain as universal numerical values. Their universality reflects the stability of the emergent structure, not an intrinsic property of physics.

Summary

All observable physical properties — spin, charge, fields, geometry, dynamics, symmetries, and constants — are effective representations of structural characteristics of the emergent domain. None of them are fundamental: they are compressed images of regularities, tensions, rigidities, and symmetries of the pattern $(X, D_S^{(e)}, C_e)_{\text{reg}}$. The projection Φ does not create these properties, but translates them into physical forms consistent with the emergent geometry.

7. General properties of the emergent physical domain

The emergent physical domain is neither an extension of the fundamental domain nor a reconstruction of it. It is an effective representation of the pregeometric structure $(X, D_S^{(e)}, C_e)_{\text{reg}}$ obtained through the projection map Φ . Physical properties are not primitive: they emerge as compressed images of the emergent correlation and the emergent structural distance. This section describes the general features of the physical domain that follow inevitably from the ontological hierarchy of the PSE.

7.1 Non-fundamentality of physical quantities

In the physical domain, no quantity is fundamental. Distance, metric, time, dynamics, fields, statistical distributions, symmetries, and energy are effective representations of the emergent structure $(D_S^{(e)}, C_e)$.

Physical constants are not primitive: they are images of structural parameters of the fundamental domain (such as the structural limit c).

The physical domain contains no original information: it is a monotonic compression of the emergent structure and cannot reconstruct the higher levels of the hierarchy.

7.2 Emergent locality

Physical locality belongs to the emergent domain and appears when the structure $(D_S^{(e)}, C_e)$ reaches a level of complexity capable of generating stable and coherent configurations. The projection map Φ

represents this organization within the physical domain, translating structural order into geometric order.

Features of Emergent Locality

- Physical locality expresses the organization of the emergent structure.
- Its presence depends on the complexity achieved by the emergent domain.
- Its stability varies across different classes of Φ -maps and across distinct emergent configurations.
- Phenomena such as entanglement show that physical distance may diverge from structural proximity, revealing correlations that traverse the geometric ordering.

Entanglement

Entanglement arises from structural proximity within $(D_S^{(e)}, C_e)$: physically separated points can share strong correlations when they occupy contiguous regions in the emergent structure.

7.3 Emergent causality

Physical causality belongs to the emergent domain and appears when the structure $(D_S^{(e)}, C_e)$ reaches a level of complexity sufficient to define a stable causal order. The projection Φ represents, in the physical domain, the geometric and dynamical form taken by the emergent structure, making visible:

- **an effective temporal ordering,**
- **a causal structure C** consistent with the emergent organization,
- **a propagation limit,** the physical image of the structural limit c .

Physical causality is a derived phenomenon, rooted in the configuration of the emergent domain and in its representation through Φ . Its presence and specific form depend on the class of emergent structures and on their corresponding physical representations.

7.4 Physical time as an effective representation

Physical time belongs to the emergent domain and takes shape when the structure $(D_S^{(e)}, C_e)$ reaches a level of complexity capable of supporting a stable temporal order. The structural index t identifies the fundamental configurations of the prespatial structure; this organization becomes physical time when the projection Φ represents it in the physical domain as a temporal coordinate or an evolution parameter.

Physical time may acquire geometric or dynamical features, such as:

- a continuous temporal coordinate,
- an evolution parameter associated with dynamics,
- an effective causal structure.

Physical time thus derives from the structural organization expressed by t and from its representation through Φ , acquiring metric, continuous, or reversible properties when the complexity of the emergent domain supports them.

7.5 Physical dynamics as compression

Physical dynamics is an effective representation of coherent variations in the emergent correlation. The projection Φ expresses these variations as physical evolution, producing equations of motion that compress structural regularities into a physical form.

The equations of motion reflect:

- structural regularities of the emergent domain,
- properties specific to the generated physical representation,
- informational limits of the projection.

Physical dynamics belongs to the emergent domain and takes the form supported by the complexity of the structure $(D_S^{(e)}, C_e)$.

7.6. Emergent Symmetries

Physical symmetries describe the organization of laws, the structure of interactions, and the stability of phenomena. In the PSE they belong to the emergent domain and arise from the global properties of the structure $(X, D_S^{(e)}, C_e)_{\text{reg}}$. The projection Φ represents these properties in the physical domain, formalizing them as geometric, dynamical, and discrete invariants.

Symmetries originate from the mathematical constraints of the emergent structure: continuity, isotropy, saturation, stability, and coherence. The representation Φ expresses these features in physical form, making symmetries visible as properties of the compressed image of the emergent correlation.

7.6.1 Structural origin of symmetries

The emergent structure is defined by two continuous functions:

- the emergent structural distance $D_S^{(e)}$,
- the emergent correlation C_e ,

linked by a monotonically decreasing constraint:

$$C_e(x, y) = F\left(D_S^{(e)}(x, y)\right), \quad F' < 0.$$

This relation establishes that:

- the correlation reflects the full information contained in the distance,
- the distance determines the global form of the emergent structure,
- the representation Φ inherits continuity, stability, and monotonicity.

Physical symmetries therefore express global properties of the continuous and isotropic structure $(D_S^{(e)}, C_e)$.

7.6.2 Lorentz symmetry as a property of the emergent metric

The continuity and regularity of the emergent structural distance are reflected in the geometric form represented by Φ . A continuous and isotropic structure without privileged directions produces a physical representation consistent with regular linear transformations.

From this organization arise:

- invariance under regular linear transformations,
- isotropy of the emergent domain,
- absence of privileged geometric scales.

Lorentz symmetry originates from the regularity and isotropy of the emergent structure and is formalized by Φ in the physical metric.

7.6.3 Gauge symmetries as a property of the emergent correlation

The emergent correlation C_e is:

- continuous,
- non-local,
- compressive,
- monotonic,
- determined by $D_S^{(e)}$.

These features imply that the correlation is defined up to transformations that preserve the functional constraint linking distance and correlation.

Gauge symmetries represent transformations that:

- preserve the emergent correlation,
- maintain the compressive character of the representation Φ ,
- leave physical quantities invariant.

The compressed correlational structure expressed by Φ formalizes gauge symmetries in the physical domain.

7.6.4 Invariance of physical laws

Emergent physical laws retain their form under transformations that preserve the emergent structure. Since Φ is monotonic and compressive, physical invariance appears with respect to:

- transformations preserving the emergent metric,
- transformations preserving the emergent correlation,
- transformations preserving compressivity.

Physical invariance thus derives from the global coherence of the emergent structure and from its representation in the physical domain.

7.6.5 Stability of symmetries

The stability of emergent physical fields reflects the global coherence of the structure $(D_S^{(e)}, C_e)$. Symmetries maintain their form under local variations of the emergent structure, showing that:

- Lorentz symmetry is stable,
- gauge symmetries are stable,

- discrete symmetries are stable.

The stability of symmetries belongs to the emergent domain and arises from the continuity and coherence of the structure represented by Φ .

7.6.6 Summary

Physical symmetries arise from the structure $(D_S^{(e)}, C_e)$. Φ represents them as invariants of the emergent form. Their origin is structural, their form is geometric, their stability is global.

7.7. Emergent Physical Constants

Fundamental physical constants — the correlation scale $\hbar$, the structural limit c , and the compressivity coefficient G — belong to the emergent domain. In the PSE they arise from the global properties of the pregeometric structure and from the form taken by the emergent correlation $(D_S^{(e)}, C_e)$. The projection Φ represents these properties in the physical domain, translating them into universal constants of effective physical theories.

Physical constants emerge from three structural elements of the fundamental domain:

- **discreteness of the prespatial domain,**
- **binary nature of the fundamental correlation,**
- **structural threshold Δ** selecting admissible correlations.

These elements define minimal scales, global limits, and response coefficients that Φ represents as physical constants.

7.7.1 Structural origin of physical constants

The ontological chain of the PSE:

$$S \rightarrow D_S^{(f)} \rightarrow C_f \rightarrow D_S^{(e)} \rightarrow C_e \rightarrow \Phi \rightarrow \mathcal{D}_{\text{physical}}$$

organizes the emergent structure through three fundamental steps:

1. Discreteness of the fundamental domain

The prespatial structure is discrete and introduces:

- ontological granularity,
- a minimal structural scale,
- original non-continuity.

This granularity becomes the basis of minimal physical scales.

2. Binary correlation C_f

The fundamental correlation takes discrete values:

$$C_f(i, j) \in \{0, 1\},$$

and introduces:

- structural quantization,
- discrete selectivity,

- informational rigidity.

This selectivity is the root of physical quantization.

3. Structural threshold Δ

The threshold Δ selects the correlations that contribute to the emergent structure, producing:

- saturation,
- regularity,
- global coherence.

Δ determines which regions of the fundamental domain participate in the emergent form.

Conceptual result

These three elements generate three universal scales:

- $\hbar$ as the minimal scale of emergent correlation,
- c as the upper limit of admissible structural reorganization,
- G as the compressivity coefficient of the physical representation.

Physical constants belong to the emergent structure and are represented by Φ as universal parameters of physical theories.

7.7.2 Emergence of the $\hbar$ scale

The $\hbar$ scale expresses the minimal variation of the emergent correlation. The binarity of C_f and the selectivity imposed by Δ produce a granularity that survives collective regularization, generating a minimal variation of the correlation:

$$\hbar_{\text{structural}} = \min \Delta C_e.$$

This scale represents:

- the minimal granularity of the emergent correlation,
- the structural basis of quantization,
- the informational threshold of physical variations.

Φ represents this scale as the Planck constant in the physical domain.

7.7.3 Emergence of the structural limit c

The limit c expresses the maximal amplitude of admissible reorganizations of the emergent structure. The emergent distance $D_S^{(e)}$ possesses a global saturation that defines an upper bound for structural variations.

The representation Φ translates this bound into physical form as:

- the maximal speed of observable variations,
- the limit of the emergent metric,
- the relativistic constant of effective theories.

c belongs to the emergent structure and is represented by Φ as the physical speed limit.

7.7.4 Emergence of the coefficient G

The coefficient G expresses the response of the physical representation to the compressivity of the emergent structure. The projection Φ compresses the structure $(D_S^{(e)}, C_e)$ into the physical domain, producing:

- curvature as a compressed image of the emergent distance,
- mass as a compressed image of saturated regions,
- a mass–curvature relation determined by the degree of compression.

G represents the response coefficient of the physical metric to structural compression.

7.7.5 Summary

Physical constants arise from the structure $(D_S^{(e)}, C_e)$. Φ represents them as universal scales of the emergent form. $\hbar$ expresses minimal granularity, c the structural limit, G the compressive response.

7.8 Non-Invertibility of the Physical Domain

The physical domain contains less information than the emergent domain. Consequently:

- no physical quantity can reconstruct $D_S^{(e)}$;
- no physical structure can reconstruct $D_S^{(f)}$;
- no physical dynamics can influence the fundamental structure.

Physics is ontologically non-retroactive. The physical domain is a compressed representation, not a fundamental level.

8. Connection with Operational Physics

Operational physics describes the physical domain through metrics, fields, dynamics, and discrete quantities. In the PSE these structures are not fundamental: they belong to the emergent domain and are represented by Φ as compressive images of the structure $(D_S^{(e)}, C_e)$.

Known physical theories — general relativity, quantum mechanics, field theory, and classical dynamics — correspond to different regimes of the same emergent structure. Their mutual consistency is not imposed: it follows from the global coherence of the emergent form.

8.1 Emergence of General Relativity (GR)

General relativity treats the metric and curvature as fundamental elements. In the PSE, both metric and curvature belong to the emergent domain and are represented by Φ as geometric forms of the structure $(D_S^{(e)}, C_e)$.

Conceptual origin

The emergent distance is:

- continuous,
- regular,
- isotropic,

- saturating.

The representation Φ expresses these properties as a physical metric that is coherent and free of privileged directions.

Operational regime

GR emerges as an effective limit when:

- compressivity is weak,
- the emergent structure is nearly isotropic,
- the emergent correlation is stable.

In this regime, the metric represented by Φ reproduces:

- geodesics,
- regular curvature,
- tensorial structures,
- geometric dynamics.

General relativity is therefore an effective representation of the emergent form.

8.2 Emergence of Quantum Mechanics (QM)

Quantum mechanics is characterized by nonlocality, minimal granularity, and non-classical correlations. In the PSE these properties arise from the form taken by the emergent correlation.

Conceptual origin

The emergent correlation C_e :

- is continuous,
- is nonlocal,
- is compressive,
- derives from a binary correlation.

Collective regularization preserves a minimal variation of the correlation, defining a structural scale:

$$\hbar_{\text{structural}} = \min \Delta C_e.$$

This scale represents the minimal granularity of the emergent correlation.

Operational regime

QM emerges as an effective limit when:

- compressivity is strong,
- the emergent structure is highly correlated,
- the minimal variation of the correlation dominates the dynamics.

In this regime Φ represents:

- nonlocality,
- minimal granularity,
- action thresholds,
- state spaces consistent with the emergent structure.

8.3 Emergence of Quantum Field Theory (QFT)

Quantum field theory describes continuous fields, operators, and propagators. In the PSE, fields arise as compressive images of the emergent correlation.

Conceptual origin

The emergent correlation is:

- continuous,
- regular,
- saturating.

The compressive nature of Φ produces physical fields that are:

- continuous,
- regular,
- free of uncontrolled divergences.

Operational regime

QFT emerges as an effective limit when:

- the emergent correlation is quasi-local,
- the emergent metric is regular,
- compressivity is moderate.

In this regime the fields represented by Φ reproduce regular propagators and effective local interactions.

8.4 Emergence of Classical Dynamics

Classical dynamics describes trajectories and equations of motion. In the PSE, physical dynamics is a compressive representation of coherent variations in the emergent structure.

Conceptual origin

The emergent correlation and the emergent metric define a geometric and dynamical order that Φ expresses as physical evolution.

Operational regime

Classical dynamics emerges as an effective limit when:

- the emergent correlation is weak,
- the emergent metric is nearly flat,

- compressivity is minimal.

In this regime the equations of motion represented by Φ coincide with classical ones.

8.5 Unity of Operational Theories

Operational physical theories are not independent: they represent different ways in which the emergent structure is expressed by Φ . Their compatibility follows from the global coherence of the form $(D_S^{(e)}, C_e)$. Each physical theory corresponds to a specific regime of correlation, saturation, and compressivity.

8.6 Summary

The physical domain is a compressive representation of the emergent structure. Φ does not generate physics: it formalizes it as a coherent image of the pregeometric form. GR, QM, QFT, and classical dynamics are different regimes of the same structure $(D_S^{(e)}, C_e)$. Their unity follows from the ontological coherence of the PSE.

9. Synthesis of the Conceptual Framework

The Conceptual Framework of the PSE establishes that fundamental reality is not physical, geometric, or dynamical, but structural. The prespatial structure S is ontologically real while remaining unobservable and lacking coordinates, metric, causality, or dynamics. The relations among the elements of S are reorganized by the fundamental process $\mathfrak{D}$, which generates the discrete fundamental structural distance $D_S^{(f)}$, responsible for the internal coherence of the structure. From $D_S^{(f)}$, through the ontological threshold Δ , derives the fundamental correlation C_f , binary and selective.

The aggregation of fundamental correlations produces the continuous emergent structural distance $D_S^{(e)}$ and the emergent correlation C_e , both unobservable yet already regular. These quantities constitute the emergent domain X , a pregeometric level containing continuous relational information that is not yet physical.

Physical quantities emerge only when the emergent structure reaches a sufficient level of complexity. The monotonic projection

$$\Phi: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow \mathcal{D}_{\text{phys}}$$

compresses the pregeometric structure into effective representations: geometry, dynamics, fields, statistical distributions, discrete structures, and quantum structures. Φ does not introduce any autonomous physicality: it formalizes the portion of the emergent structure that has already acquired enough complexity, stability, and coherence to be represented as physical quantity.

Each level of the ontological chain

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{\Phi} \mathcal{D}_{\text{phys}}$$

contains less information than the previous one and is non-retroactive: no physical quantity can reconstruct or influence the fundamental structure. Observability is an emergent property that appears only in the physical domain formalized by Φ .

Fundamental reality also gives rise to emergent physical constants: the threshold Δ and structural discreteness imply a minimum correlation scale that appears in the physical domain as $\hbar$; the

structural limit c becomes, in the physical representation, the relativistic bound; the compressivity of Φ produces a response coefficient that in the physical domain takes the form of the gravitational constant G . None of these constants is fundamental: all are compressed images of structural constraints of the emergent domain.

The Conceptual Framework shows that observable physics is not a set of primitive axioms, but the compressed manifestation of a real prespatial structure. Geometry, dynamics, causality, fields, and quantum correlations emerge as effective representations of structural correlation, providing a unified ontological basis for the emergence of physical quantities and for the compatibility between local and non-local phenomena.

Part III - Theoretical Positioning

The PSE arises from a methodological observation: a theory of physical emergence can be coherent only if the ontological level contains no physical concept.

From this premise follows a negative analysis: by eliminating all structures that introduce physics into the ontological level, what remains is a minimal chain of non-physical, non-geometric, non-dynamical, and non-causal elements. This chain is not a creative choice but the result of an exclusion process: what remains when everything that generates incoherence is removed.

The PSE is therefore not a speculative proposal, but an inevitable structure for anyone who wishes to describe physical emergence from a non-physical source without incurring circularity or implicit assumptions.

1. Unresolved Problems in Emergentist Physics

1.1 Geometric Circularity

Many emergentist approaches attempt to derive spacetime geometry from more primitive structures. However, these structures already incorporate implicit geometric elements: nodes with adjacencies, graphs with connections, networks with discrete distances, correlations with underlying topologies. In such cases, geometry does not emerge: it is presupposed. The theory becomes circular because what should be derived is already present at the ontological level.

1.2 Dynamical Circularity

Similarly, several models assume dynamics at the pre-physical level: state evolutions, node updates, transition rules, unitary or stochastic operators. If dynamics is already present at the ontological level, then physical dynamics does not emerge: it is inherited. This prevents explaining the origin of physical dynamics and makes a non-circular emergence theory impossible.

1.3 Failure to Produce Discrete Quantities

Discrete physical quantities (mass, spin, charge, quantized states) do not emerge from correlations or pregeometric structures unless they are explicitly introduced. Correlations alone do not select or stabilize discrete values: they produce patterns, not quantities. To obtain discrete physical quantities, a mechanism of **compression and selection** is required—absent in all known emergentist theories. This leaves unresolved one of the central problems of fundamental physics.

1.4 Incompatibility Among Emergent Physical Domains

Emergentist theories tend to generate a single domain: geometry, or dynamics, or discrete structures. There is no framework that simultaneously generates:

- geometry,
- discrete quantities,
- dynamics,
- stable configurations,

from a single source. The lack of multi-emergence leads to internal incompatibilities: geometry and particles do not derive from the same structure, and dynamics is not coherent with discrete quantities.

1.5 Absence of a Coherent Non-Physical Source

Many models assume implicit physical concepts at the ontological level: causality, nodes, states, topologies, metrics, continuity or discreteness. This prevents defining a truly non-physical source. A theory of physical emergence requires that the ontological level be free of any physical concept; otherwise, emergence becomes impossible or circular.

1.6 Dependence on Implicit Physical Structures

Even when not explicitly declared, many theories incorporate:

- temporal succession,
- evolution,
- causal relations,
- notions of distance,
- topological constraints.

If these elements are present at the ontological level, their emergence cannot be explained. A coherent theory must avoid any form of implicit physics in the pre-physical level.

2. Negative Analysis: What Cannot Work

Physical emergence requires that the ontological level be devoid of any physical concept. This principle, seemingly simple, has drastic consequences: it eliminates almost all the structures used in contemporary emergentist theories. The present analysis does not yet introduce the PSE: it shows what **cannot** constitute a coherent ontological level.

2.1 Why Networks Cannot Generate Geometry

Many pre-geometric models assume networks, graphs, or discrete structures as an ontological basis. However, such structures inevitably incorporate:

- adjacencies,
- connections,
- relations of proximity,
- topological constraints.

These elements are already forms of implicit geometry. If geometry is present at the ontological level, it cannot emerge: it is simply transferred. A theory of geometric emergence must therefore exclude any structure that contains spatial relations, even if discrete.

2.2 Why Non-Physical Correlations Cannot Generate Discrete Physical Quantities Without a Compression Map

Correlations—even emergent ones—do not select or stabilize discrete values. A pregeometric correlation may contain patterns, constraints, saturations, rigidity, and compatibility structures, but it does **not** contain physical quantities.

Discrete quantities (mass, spin, charge, quantized states) require a mechanism of **compression and selection** that transforms emergent constraints into discrete physical values.

Consequently, no correlation structure

$$C_f, C_e$$

can generate a discrete physical quantity without a compression map:

$$\Phi_{\text{field}}\left(X, D_S^{(e)}, C_e\right)_{\text{reg}} \rightarrow \{\text{discrete physical quantities}\}.$$

Without such a mechanism, correlations cannot produce discrete physical quantities.

The map Φ_{field} formalizes the level of complexity of the emergent structure from which physical quantities spontaneously emerge. It does not introduce physics into the ontological domain: it extracts discrete quantities only when the emergent structure has reached a regime of saturation, compatibility, and rigidity sufficient to be represented as a physical value.

2.3 Why Dynamics Cannot Be Fundamental

Any form of dynamics at the ontological level inevitably introduces:

- succession,
- change,
- evolution,
- causality,
- implicit time.

These elements are physical concepts. If dynamics is fundamental, then physical dynamics does not emerge: it is inherited from the ontological level. A theory of emergence must therefore exclude any form of ontological evolution, even when not interpreted as physical time.

In particular, no ontological rule of the form

$$X \rightarrow X'$$

can be admitted, because it already encodes temporal succession—i.e., an implicit physical order.

2.4 Why Causality Cannot Be Assumed

Causality is a physical relation. Assuming it at the ontological level means introducing:

- directionality,
- order,
- succession,
- temporal structure.

An emergentist theory that assumes causality cannot explain the emergence of physical causality. Causality must be an emergent phenomenon, not an ontological element.

Thus, no ontological relation of the form

$$x < y$$

can be present at the fundamental level.

2.5 Why No Physical Structure Can Be Present at the Ontological Level

Any physical concept introduced at the ontological level—even implicitly—prevents emergence. Among these:

- distance,
- metric,
- topology,
- continuity,
- discreteness,
- states,
- nodes,
- operators,
- evolution,
- succession.

A theory of physical emergence must be built on a completely non-physical ontological level. This eliminates almost all structures used in known emergentist theories.

In particular, the ontological level cannot contain any object of the form:

$$d(x, y), g_{\mu\nu}, \mathcal{T}, \rho, \hat{H}, \hat{U}(T),$$

nor any state space, discrete or continuous.

3. The Ontological Chain of the PSE as an Inevitable Structure

The negative analysis shows that no physical, geometric, dynamical, or causal structure can be assumed at the ontological level without generating circularity. Once all forms of physical contamination are eliminated, only one possibility remains: a chain of non-physical, non-geometric, non-dynamical, and non-causal elements, each necessary to guarantee coherence and closure.

This chain is:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}.$$

It is not a proposal: it is an inevitable deduction.

3.1 Presentation of the Chain

The ontological chain of the PSE consists of six elements, each with a precise role:

- S : Fundamental prespatial structure, real (non-geometric, non-topological, non-dynamical).
- $D_S^{(f)}$: Fundamental structural distance (non-physical).
- C_f : Fundamental correlation (non-physical), regulated by the ontological threshold Δ .
- $D_S^{(e)}$: Emergent structural distance (non-physical).

- C_e : Emergent correlation (non-physical).
- $\mathcal{D}_{\text{physical}}$: geometry, discrete quantities, dynamics, fields, statistics.

The transformations between these levels are represented by the operators $\mathfrak{D}$, F_λ , G_λ , and Φ , which are not ontological elements but structural maps. The threshold Δ is instead a fundamental ontological constraint of the prespatial domain.

Each element of the chain is necessary to avoid circularity and to guarantee the coherent emergence of physical domains.

3.2 Mathematical Foundations of the Ontological Chain

The ontological chain of the PSE is founded on a rigorous mathematical structure. The starting point is a discrete pregeometric set S , devoid of metric, causality, and dynamics, on which are defined:

- a fundamental structural distance $D_S^{(f)}$,
- a fundamental correlation C_f ,
- a structural threshold Δ , which determines the selectivity of correlations.

These objects are neither physical nor observable: they constitute the primary ontological structure.

The transition from the fundamental level to the non-observable emergent level is realized through monotonic families of regularization operators:

$$\{F_\lambda\}, \quad \{G_\lambda\},$$

which generate:

- the emergent structural distance $D_S^{(e)}$,
- the emergent correlation C_e ,

as continuous limits of the fundamental quantities.

The fundamental reorganization $\mathfrak{D}$ selects the coherent configurations of the prespatial structure S , stabilized by the structural principle $E + C$. The $E + C$ principle determines resistance to structural relaxation: it minimizes contradictions and reduces tensions among nearby relations, ensuring that only compatible and stable configurations persist in the fundamental domain.

The compression maps Φ represent the emergent structure in physical domains, producing geometry, dynamics, fields, discrete quantities, observables, and measurements as effective images of the emergent correlation.

These transitions introduce no physical structure: they generate exclusively non-physical emergent constraints, which will then be compressed by the maps Φ into observable physical domains.

3.3 Why Each Element Is Necessary

3.3.1 Ontological Elements

The Structure S

A non-physical source is required; otherwise physics would be fundamental and emergence impossible.

The Fundamental Structural Distance $D_S^{(f)}$

A fundamental distance is required; otherwise the source would lack any organization and could not generate correlations.

The Fundamental Correlation C_f

A fundamental correlation is required; otherwise no mechanism of emergence would exist.

The Emergent Structural Distance $D_S^{(e)}$

An emergent distance is required; otherwise C_f could not produce stable constraints or patterns.

The Emergent Correlation C_e

An emergent correlation is required; otherwise no constraints would exist that could be selected by the projection Φ .

3.3.2 Fundamental Ontological Constraints (Not Operators)

The Structural Index t

It is required to order the fundamental reorganization, ensuring coherence of configurations without introducing physical dynamics.

The Structural Principle $E + C$

It is required to stabilize the fundamental structure: it minimizes contradictions and tensions, ensuring that only compatible configurations persist.

The Threshold Δ

It is required to select the relevant fundamental correlations; otherwise C_f would be indeterminate and no coherent structure would emerge.

The Structural Limit c

It is required to limit admissible variations between adjacent configurations, ensuring local coherence of the fundamental reorganization.

3.3.3 Necessary Operators (Non-Ontological)

The Operator $\mathfrak{D}$

It is required to reorganize the fundamental structure S and generate $D_S^{(f)}$.

The Operator F_λ

It is required to aggregate fundamental correlations and generate the emergent distance $D_S^{(e)}$.

The Operator G_λ

It is required to regularize the emergent distance and generate the emergent correlation C_e .

The Projection Φ

It is required to compress the emergent structure into the physical domain. Φ is not an ontological element: it formalizes $(X, D_S^{(e)}, C_e)_{\text{reg}}$ into observable physical quantities.

3.3.4 Non-Ontological Result

The Physical Domains $\mathcal{D}_{\text{physic}}$

They are the final result: what we observe as geometry, discrete quantities, dynamics, fields, and statistics.

3.4 Why Nothing Can Be Removed

If any ontological element is removed:

$$\text{Chain} \setminus X \Rightarrow \text{ontological collapse.}$$

Ontological Elements

- **without** S , no ontological source exists;
- **without** $D_S^{(f)}$, S has no internal organization and cannot generate correlations;
- **without** C_f , no emergence exists;
- **without** $D_S^{(e)}$, no emergent structure exists;
- **without** C_e , no emergent constraint exists.

Fundamental Ontological Constraints (Non-Operators)

- **without** t , the fundamental reorganization cannot be ordered and the emergent structure does not stabilize;
- **without** $E + C$, no structural stabilization exists and the fundamental distance $D_S^{(f)}$ cannot emerge;
- **without** Δ , no selection of fundamental correlations exists and C_f is indeterminate;
- **without** c , no structural limit exists and the fundamental reorganization loses local coherence.

Necessary Operators (Non-Ontological)

- **without** $\mathfrak{D}$, no reorganization of the fundamental structure exists and $D_S^{(f)}$ cannot emerge;
- **without** F_λ , no aggregation of fundamental correlations exists and $D_S^{(e)}$ cannot emerge;
- **without** G_λ , no regularization of the emergent distance exists and C_e cannot emerge;
- **without** Φ , the emergent structure cannot be compressed into the physical domain and no physical quantity can emerge.

Non-Ontological Result

- **without** $\mathcal{D}_{\text{physical}}$, the theory describes nothing observable.

3.5 Why Nothing Can Be Added

Adding elements means introducing implicit physical concepts:

- dynamics $\Rightarrow$ dynamical circularity;
- geometry $\Rightarrow$ geometrical circularity;
- causality $\Rightarrow$ physical contamination;
- states $\Rightarrow$ quantum contamination;
- nodes $\Rightarrow$ topological contamination.

Formally:

$$\text{Chain} + X_{\text{physical}} \Rightarrow \text{incoherence.}$$

Any addition makes the theory arbitrary or incoherent.

3.6 Why Nothing Can Be Modified

Modifying the chain means altering essential roles:

- changing the role of correlations $\Rightarrow$ loss of emergence;
- changing the role of distances $\Rightarrow$ loss of stability;
- changing the role of maps $\Rightarrow$ loss of multi-emergence;
- changing the order $\Rightarrow$ loss of ontological closure.

The chain is the only sequence that:

does not assume physics, geometry, dynamics, causality, or quantities

and generates everything we observe.

4. The Fundamental Structural Reorganization $\mathfrak{D}$ and the Ontological Parameter t

The ontological chain of the PSE requires that the prespatial structure S not be static, but possess a property of fundamental structural reorganization, purely non-physical. This reorganization modifies the fundamental structural distance $D_S^{(f)}$ without introducing physical time, causality, or propagation: it is not an ontological dynamics, but a local property of the structure.

Fundamental configurations are indexed by an internal discrete parameter t , non-physical and non-metric, which orders the configurations of the prespatial structure without defining succession, duration, or evolution. The parameter t does not generate the reorganization: it merely provides an ontological labeling.

In this section we do not describe the fundamental reorganization $\mathfrak{D}$ (already formalized in the Conceptual Framework), but explain why t , c , and the $E + C$ principle are inevitable, and why no physical quantity can exist at the fundamental levels.

4.1 The Structural Parameter t

The structural parameter t is a discrete index that orders the fundamental configurations of the prespatial structure:

$$S_t \text{ with } t \in \mathbb{N}.$$

Its introduction is ontologically necessary:

- without an internal ordering, the fundamental structure would be indeterminate;
- such ordering cannot be physical, because no physical quantity is allowed at the fundamental level;
- it cannot be continuous, metric, or causal, otherwise it would reintroduce physical time or geometry.

For this reason, t :

- is not physical time,
- does not imply causality,

- does not define physical succession,
- does not belong to any physical domain.

t is a minimal ontological index: it serves only to make the ontological sequence of prespatial configurations well-defined. It does not generate the next configuration and does not encode any form of evolution.

Physical time emerges exclusively after the projection Φ .

4.2 The Structural Coherence Limit c

The fundamental structural reorganization is subject to an internal structural coherence limit c , which regulates the extent of the rewritings of $D_S^{(f)}$:

$$D_S^{(f)} \rightarrow D_S^{(f)} *$$

Since S is discrete and composed of elementary units with finite-support rewriting rules, each transformation can involve only a finite number of elementary units. From this, the existence of an internal limit c necessarily follows.

This limit:

- is not a physical velocity,
- does not imply causality,
- does not represent physical propagation,
- does not belong to any physical domain.

c is an internal structural constraint that prevents instantaneous global reorganizations. The relativistic limit observed in the physical domain is the compressive image of c through Φ .

4.3 The Structural Principle $E + C$

The fundamental structural reorganization is guided by a principle of structural minimization:

$$E + C = \text{minimization of contradictions} + \text{reduction of tensions.}$$

- **E : internal coherence** — eliminates incompatible configurations.
- **C : local compatibility** — reduces tensions among nearby relations.

This principle is necessary because:

- without a selection criterion, the reorganization would be arbitrary;
- the criterion cannot be physical (energy, action, equations of motion), because no physical quantity is allowed at the fundamental level.

For this reason, $E + C$:

- does not minimize energy,
- does not define equations of motion,
- does not introduce causality,
- does not imply temporal continuity.

It is a purely structural principle: the minimal form of ontological selection compatible with the non-physical nature of the fundamental level.

4.4 Nature of the Fundamental Structural Reorganization $\mathfrak{D}$

The fundamental structural reorganization $\mathfrak{D}$ acts locally on the prespatial structure S , rewriting the fundamental relations in a manner compatible with the discrete structural index t . The index t does not define any physical succession: it is an ontological parameter that **labels** the configurations of the prespatial structure, without generating the next configuration and without introducing time, causality, or evolution.

The reorganization is formalized as an internal transformation:

$$\mathfrak{D}(S_t) = S_{t^+},$$

where t^+ denotes a configuration compatible with S_t according to the structural constraints. This relation does not represent a “next step,” does not imply directionality, and does not encode any form of ontological dynamics: it is a **local rewriting** of the structure, not a process in time.

Compatibility between configurations is limited by the structural coherence parameter c , which prevents incompatible rewritings and does not introduce any notion of velocity, propagation, or physical limit. The structural principle $E + C$ stabilizes configurations by selecting those that minimize contradictions and tensions among nearby relations.

$\mathfrak{D}$ is discrete, non-continuous, non-physical, and non-observable. It is the only form of reorganization allowed at the fundamental level: introducing physical dynamics, equations of motion, or propagation would be conceptually illegitimate and would violate the non-physical nature of the ontological domain.

4.5 Emergence of $D_S^{(e)}$ and of the Emergent Correlation C_e

The emergent structural distance $D_S^{(e)}$ arises as a collective regularization of the transformations of $D_S^{(f)}$. It is not directly manipulated by the fundamental reorganization:

$$D_S^{(e)} = \mathcal{R}(D_S^{(f)}),$$

where $\mathcal{R}$ represents emergent regularization.

Characteristics of $D_S^{(e)}$:

- continuous,
- coherent,
- non-observable,
- non-physical.

The emergent correlation C_e is a monotonically decreasing function of $D_S^{(e)}$:

$$C_e(i, j) = F(D_S^{(e)}(i, j)),$$

and its variations derive exclusively from variations of $D_S^{(e)}$. The fundamental reorganization does not act on C_e : this constraint prevents reintroducing physical dynamics into the non-observable emergent level.

4.6 No Physical Dynamics in the Fundamental Levels

In the fundamental and non-observable emergent levels:

- no equations of motion exist,
- no physical causality exists,
- no physical propagation exists,
- no physical metric exists,
- no energy, mass, or gravity exist.

The fundamental structural reorganization is purely structural. Physics arises only after the projection Φ , when $D_S^{(e)}$ and C_e are compressed into physical quantities.

5. The Φ Maps as Relations, Not Entities

The Φ maps are one of the conceptually most delicate elements of the PSE. To avoid metaphysical, informational, or operator-like interpretations, it is essential to clarify that Φ is **not** an ontological object, **not** a dynamics, **not** a physical structure, and **not** an operator. Φ is a *class of reading relations* that formalize the portion of the emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}$ sufficiently complex to be representable as physical quantities.

The ontological level of the PSE contains exclusively:

$$S, D_S^{(f)}, C_f, D_S^{(e)}, C_e,$$

all non-physical, non-geometric, non-dynamical, and non-causal elements. The Φ maps do **not** belong to this level: they are introduced solely to enable the compression and effective representation of emergent constraints within the observable physical domain.

Formally:

$$\Phi: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow \text{Physical domains.}$$

Φ does not create physics, does not introduce observability, and does not add ontological content: it represents in physical form what has already emerged in the pregeometric domain.

5.1 Why Φ Is Not an Ontological Object

The ontological level of the PSE is closed and complete. It contains no operators, functions, transformations, or physical structures. The ontological chain:

$$S \rightarrow D_S^{(f)} \rightarrow C_f \rightarrow D_S^{(e)} \rightarrow C_e$$

does not include Φ . Φ does not belong to the chain: it is a *relation between levels*, not an element of the chain.

5.2 Why Φ Is Not Dynamics

A dynamics requires succession, evolution, causality, and time. None of these properties is present in Φ . The maps do not evolve, do not update, and do not transform over time. They are static relations:

$$\Phi(x) = y,$$

with no ontological temporal parameter.

Physical time emerges only as the image of:

$$\Phi_{\text{dyn}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (\hat{H}, \hat{U}(T), \rho),$$

where T is *physical* time, not ontological.

5.3 Why Φ Is Not Geometric

The Φ maps contain no distances, metrics, topologies, nodes, continuity, or discreteness. Geometry emerges only as the image of:

$$\Phi_{\text{geom}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (d, g_{\mu\nu}, \mathcal{C}),$$

and is not present in the domain of Φ . Φ is not a geometric structure: it is a relation that *produces* geometry.

5.4 Why Φ Is Not Physical

The Φ maps are not physical operators, dynamical transformations, evolutions, states, or processes. They do not belong to the physical domain and cannot be interpreted as physical entities.

They are reading relations that allow non-physical emergent constraints to be interpreted as coherent physical structures.

In other words:

Φ is not physics; Φ produces physics.

5.5 Role of the Classes of Maps

The classes of Φ maps have distinct roles, but none of them is ontological:

- **Geometric compression:** Φ_{geom} formalizes geometric structures.
- **Dynamical representation:** Φ_{dyn} formalizes physical dynamics.
- **Statistical extraction:** Φ_{stat} formalizes distributions and mean values.
- **Field representation:** Φ_{field} formalizes effective physical fields.
- **Discrete selection:** Φ_{disc} formalizes discrete quantities and lattice-like structures.
- **Quantum compression:** Φ_{quant} formalizes the state space $\mathcal{H}$, the action A , and the minimal threshold $\hbar$.

These roles are not ontological properties: they are *modes of reading* emergent constraints.

6. Multi-Emergence as the Unique Property of the PSE

A theory of physical emergence is coherent only if the non-physical source is capable of generating *all* observable physical domains. Most emergentist theories produce a single domain (geometry, or dynamics, or discrete structures), but cannot generate simultaneously:

- geometry,
- discrete quantities,

- dynamics,
- fields,
- statistical distributions,
- stable configurations.

The PSE is the only conceptual framework that guarantees this property, thanks to the emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}$ and to the classes of Φ maps.

Formally:

$$(X, D_S^{(e)}, C_e)_{\text{reg}} \xrightarrow{\Phi} \{\text{Geometry, Dynamics, Statistics, Fields, Discrete Structures, Quantum structure}\}.$$

6.1 Emergence of Geometry

Physical geometry emerges through:

$$\Phi_{\text{geom}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (d, g_{\mu\nu}, \mathcal{C}).$$

This map compresses emergent constraints into:

- distance d ,
- metric $g_{\mu\nu}$,
- causal structure $\mathcal{C}$.

Geometry is not present at the ontological level; it is a compressed image of the emergent structure.

6.2 Emergence of Dynamics

Physical dynamics emerges through:

$$\Phi_{\text{dyn}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (\hat{H}, \hat{U}(T), \rho).$$

This map represents compatible constraints as:

- effective Hamiltonian $\hat{H}$,
- unitary evolution $\hat{U}(T)$,
- physical state ρ .

The time T is *physical*, not ontological. Dynamics is not fundamental: it is an effective representation.

6.3 Emergence of Statistics

Statistical properties emerge through:

$$\Phi_{\text{stat}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (p(x), S, \langle O \rangle).$$

This map extracts:

- distributions $p(x)$,

- entropies S ,
- expectation values $\langle O \rangle$.

Statistics is not fundamental: it is an informational extraction of the emergent structure.

6.4 Emergence of Physical Fields

Physical fields emerge through:

$$\Phi_{\text{field}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (\phi(x), A_\mu(x), F_{\mu\nu}).$$

This map produces:

- scalar fields $\phi(x)$,
- vector fields $A_\mu(x)$,
- field tensors $F_{\mu\nu}$.

Fields are not fundamental: they are effective representations of emergent constraints.

6.5 Emergence of Discrete Structures

Discrete structures emerge through:

$$\Phi_{\text{disc}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (G, \Gamma, \mathcal{N}).$$

This map selects:

- graphs G ,
- networks Γ ,
- discrete quantities $\mathcal{N}$.

Discrete quantities are not fundamental: they are selections of saturated constraints.

6.6 Emergence of quantum structure

Quantum structure emerges through:

$$\Phi_{\text{quant}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (\mathcal{H}, A, \hbar)$$

This map compresses emergent constraints into:

- the state space $\mathcal{H}$,
- the continuous action A ,
- the minimal threshold $\hbar$.

Quantum structure is not present at the ontological level; it is an effective representation of the emergent structure.

6.7 Internal Compatibility Among Domains

Since all Φ maps share the same domain:

$$\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}$$

the physical domains emerge from a single source. This guarantees:

- compatibility between geometry and dynamics,
- compatibility between fields and discrete quantities,
- compatibility between statistics and configurations,
- absence of contradictions among domains.

In other words:

All physical domains are images of the same emergent structure. This property is unique to the PSE.

7. Structural Simplicity: The PSE as a Minimal Theory

The ontological chain of the PSE:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}$$

is not an arbitrary construction. It is the **minimal structure** that allows physical emergence without introducing circularity, physical contamination, or implicit assumptions.

In this section we show that:

1. the PSE is the simplest possible theory;
2. any simpler theory is incoherent;
3. any more complex theory is arbitrary;
4. the PSE does not assume any physical concept at the ontological level;
5. the PSE does not assume geometry, dynamics, causality, topology, or discrete quantities.

7.1 Why the PSE Is the Simplest Possible Theory

The simplicity of the PSE derives from a methodological principle:

No physical concept can be present at the ontological level.

This eliminates:

- space,
- time,
- causality,
- dynamics,
- states,
- nodes,
- metrics,

- topologies,
- discrete quantities.

Once all these elements are removed, only one structure remains: **the ontological chain of the PSE.**

There is no shorter chain that is coherent.

7.2 Why Any Simpler Theory Is Incoherent

If one attempts to simplify the chain by removing any of its elements, one obtains:

$$\text{Chain} \setminus X \Rightarrow \text{ontological collapse.}$$

Examples:

- Removing $D_S^{(f)} \rightarrow$ the fundamental structure S cannot generate correlations.
- Removing $C_f \rightarrow$ no emergence exists.
- Removing $D_S^{(e)} \rightarrow$ no stable constraints exist.
- Removing $C_e \rightarrow$ no selectable constraints exist.
- Removing $\Phi \rightarrow$ no physics exists.

Every attempt at simplification leads to a theory that cannot emerge.

7.3 Why Any More Complex Theory Is Arbitrary

Adding elements to the chain means introducing implicit physical concepts:

- adding dynamics $\rightarrow$ introduces time and causality;
- adding geometry $\rightarrow$ introduces distance and metric;
- adding nodes $\rightarrow$ introduces topology;
- adding states $\rightarrow$ introduces quantum structure;
- adding operators $\rightarrow$ introduces physical dynamics.

Formally:

$$\text{Chain} + X_{\text{physical}} \Rightarrow \text{incoherence.}$$

Any addition makes the theory:

- arbitrary,
- non-minimal,
- unnecessary,
- non-assessable.

The PSE is the only theory that contains nothing superfluous.

7.4 Why the PSE Assumes Nothing Physical

The ontological level of the PSE is composed of:

$$S, D_S^{(f)}, C_f, D_S^{(e)}, C_e,$$

all elements that are:

- non-physical,
- non-geometric,
- non-dynamical,
- non-causal,
- non-topological,
- non-discrete,
- non-continuous.

This guarantees that:

Physics emerges; it is not assumed.

7.5 Why the PSE Does Not Assume Geometry, Dynamics, Causality, or Topology

Geometry emerges through:

$$\Phi_{\text{geom}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (d, g_{\mu\nu}, \mathcal{C}),$$

and is not present at the ontological level.

Dynamics emerges through:

$$\Phi_{\text{dyn}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (\hat{H}, \hat{U}(T), \rho),$$

and is not present at the ontological level.

Causality emerges through:

$$\Phi_{\text{geom}} \rightarrow \mathcal{C},$$

and is not present at the ontological level.

Topology emerges through:

$$\Phi_{\text{disc}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (G, \Gamma),$$

and is not present at the ontological level.

In summary:

No physical structure is fundamental.

8. Technical Comparison with Alternatives

This section does not aim to evaluate or criticize existing emergentist theories, but to show—technically and without polemics—why none of them resolves the structural problems identified in Section 2. The comparison is based exclusively on the criteria of:

- geometrical circularity,

- dynamical circularity,
- absence of emergent discrete quantities,
- incompatibility among domains,
- presence of physical concepts at the ontological level.

The PSE does not arise as an alternative: it arises as **what remains** once all forms of incoherence are eliminated.

8.1 Causal Set Theory

Causal Set Theory assumes as fundamental:

- discrete elements,
- causal relations,
- partial order.

Formally, the ontological level contains:

$$\{x_i\}, x_i < x_j.$$

Problems relative to the PSE criteria

1. **Fundamental causality.** The relation $<$ is an implicit physical structure. Causality cannot emerge if it is assumed at the ontological level.
2. **Implicit geometry.** The causal order determines the Lorentzian metric (Malament's theorem). Geometry does not emerge: it is presupposed.
3. **Absence of multi-emergence.** The theory generates causality and discrete structure, but not physical discrete quantities, fields, dynamics, or statistics.

8.2 Spin Networks / Loop Quantum Gravity

The ontological level contains:

$$\Gamma, j_e, i_v,$$

where Γ is a graph, j_e are spins, and i_v are intertwiners.

Problems relative to the PSE criteria

1. **Fundamental discrete structure.** The graph Γ is an implicit geometric structure (adjacencies, nodes, connections).
2. **Fundamental discrete quantities.** The spins j_e are physical quantities assumed at the ontological level.
3. **Presupposed geometry.** Areas and volumes emerge from operators built on Γ . Geometry does not emerge from a non-physical source.
4. **Absence of multi-emergence.** The theory produces quantized geometry, but not emergent dynamics, emergent fields, or emergent statistics.

8.3 Tensor Networks / MERA / PEPS

The ontological level contains:

$\{T_i\}$, connections among tensors.

Problems relative to the PSE criteria

1. **Implicit lattice structure.** The tensor network is a discrete geometric structure.
2. **Fundamental operators.** Tensors are physical operators, not non-physical elements.
3. **Implicit dynamics.** Networks are constructed to represent already dynamical physical states.
4. **Absence of multi-emergence.** Networks generate states and correlations, but not physical discrete quantities, fields, or non-presupposed emergent geometry.

8.4 Emergent Gravity (Jacobson, Verlinde, Padmanabhan)

The ontological level contains:

- entropy,
- information,
- thermodynamics,
- microscopic degrees of freedom.

Problems relative to the PSE criteria

1. **Fundamental physical concepts.** Entropy, temperature, and energy are physical concepts assumed at the ontological level.
2. **Implicit geometry.** The metric is reconstructed from thermodynamic relations that presuppose causality and horizons.
3. **Absence of multi-emergence.** The theory generates geometry, but not discrete quantities, fields, or emergent dynamics.

8.5 Informational Models (Wheeler, Lloyd, Relational Rovelli)

The ontological level contains:

- bits,
- qubits,
- operators,
- states,
- informational relations.

Problems relative to the PSE criteria

1. **Fundamental operators and states.** Operators encode implicit physical dynamics.
2. **Presupposed discrete or quantum structure.** Bits and qubits are physical discrete quantities.

3. **Absence of multi-emergence.** The theory generates information, but not non-presupposed emergent geometry, nor fields, nor emergent dynamics.

9. New Interpretive Keys from the PSE

The PSE is not a physical theory: it is an ontological structure that generates physics as an emergent image. This section presents several interpretive keys showing how fundamental physical concepts can be reinterpreted as emergent configurations of the pregeometric structure, without introducing any physical quantity at the ontological level.

All definitions below are formulated exclusively in terms of:

- emergent structural distance $D_S^{(e)}$,
- emergent correlation C_e ,
- structural principle $E + C$,
- compression maps Φ .

9.1 Elementary Particles

An elementary particle is an emergent configuration of the structure S that maintains local stability despite the structural relaxation imposed by the $E + C$ principle. It is a pattern of correlations that:

- remains coherent in emergent time,
- resists the reduction of local tensions,
- maintains sufficient structural rigidity not to dissolve into emergent decoherence.

Particles are characterized by emergent invariants (mass, spin, charge), which are not intrinsic properties but properties of the pattern's shape and its ability to maintain coherence under structural relaxation.

Particles are not fundamental entities: they are stable configurations of the emergent structure, forms that persist because they resist the relaxation imposed by $E + C$.

9.2 Mass

Mass is an emergent rigidity of the correlation pattern: it measures how strongly a configuration resists the structural relaxation imposed by the $E + C$ principle.

It is the amount of structural coherence required to keep the pattern stable in emergent time, opposing the reduction of local tensions and preserving its shape in $D_S^{(e)}$ and C_e .

It is not a fundamental property: it is a property of the pattern's shape, reflecting its ability to maintain coherence under structural relaxation.

Formally:

$$\text{mass} = \text{emergent rigidity of the pattern with respect to } (E + C).$$

9.3 Energy

Energy is an emergent resistance to the structural relaxation of the pattern, determined by the $E + C$ principle. It reflects the degree to which an emergent configuration:

- maintains its form,
- resists the reduction of local tensions,
- preserves internal coherence despite the relaxation imposed by $E + C$.

It is an emergent property of the pregeometric structure: it is not a fundamental quantity and not a physical magnitude at the ontological level.

In the physical domain, energy appears as a measurable quantity because it is effectively represented by dynamical maps, but its origin is purely structural.

9.4 Spin

Spin is an internal symmetry of the shape of the emergent pattern. It is a structural invariant that remains stable under the relaxation imposed by the $E + C$ principle: a property of the configuration that is not altered by the reduction of local tensions.

It is not a physical rotation: it is an emergent feature of the pattern's shape, persisting as an internal symmetry in $D_S^{(e)}$ and C_e .

In the physical domain, spin appears as a discrete quantity because its stability is selected by the discrete maps, but its origin is purely structural.

9.5 Charge

Charge is an internal asymmetry of the shape of the emergent pattern. It is a structural invariant that remains stable under the relaxation imposed by the $E + C$ principle: a feature of the configuration that is not eliminated by the reduction of local tensions.

It is not a fundamental property: it is an emergent property of the pattern's shape, persisting as an internal asymmetry in $D_S^{(e)}$ and C_e .

9.6 Physical Fields

Physical fields are effective representations of emergent constraints of the pregeometric structure. They are continuous images of discrete configurations in $D_S^{(e)}$ and C_e : they describe how emergent coherence varies when read in the physical domain.

They are not fundamental entities: they are continuous forms that represent, in the physical domain, structural constraints that exist in the network of emergent correlations.

Field maps do not generate fields: they represent them as effective functions, translating emergent constraints into observable physical structures.

9.7 Geometry

Geometry is an effective compression of the emergent structure. It is the physical image of the constraints present in $D_S^{(e)}$ and C_e , represented as:

$$\Phi_{\text{geom}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (d, g_{\mu\nu}, \mathcal{C}).$$

It is not present at the ontological level: it is not a fundamental distance, not a metric, not a topology. It is a continuous representation of emergent correlation, obtained by compressing the pregeometric structure into effective geometric forms.

Geometry is not a fundamental physical entity: it is an image of the emergent structure in the physical domain.

9.8 Dynamics

Dynamics is an effective representation of emergent constraints of the pregeometric structure. It describes how emergent coherence is read in the physical domain when compressed into temporal and operatorial forms through:

$$\Phi_{\text{dyn}}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (\hat{H}, \hat{U}(T), \rho).$$

The time T is not ontological: it is a physical parametrization of the dynamical reading, introduced by the effective representation and not present in the fundamental structure.

Dynamics is not a fundamental property: it is a physical image of structural relaxation and emergent constraints, translated into operators, evolutions, and states in the physical domain.

10. Synthesis of the Theoretical Positioning

The Theoretical Positioning of the PSE shows that a theory of physical emergence can be coherent only if the ontological level is completely free of physical concepts. The analysis of contemporary emergentist models highlights five structural obstacles: geometrical circularity, dynamical circularity, lack of emergent discrete quantities, incompatibility among emergent domains, and the presence of implicit physical concepts in the pre-physical level. These problems are not technical details: they prevent the construction of a complete framework for physical emergence.

The negative analysis of the PSE eliminates every structure that introduces physics at the ontological level. Once every form of node, state, metric, topology, causality, succession, dynamics, or discrete quantity is removed, only one chain remains: a minimal sequence of non-physical, non-geometric, non-dynamical, and non-causal elements:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}.$$

Each element is necessary:

- without S , no source exists;
- without $D_S^{(f)}$, no organization exists;
- without C_f , no emergence exists;
- without $D_S^{(e)}$, no emergent continuity exists;
- without C_e , no selectable constraints exist;
- without Φ , no physics exists;
- without $\mathcal{D}_{\text{physical}}$, no observability exists.

Likewise, nothing can be added: any addition reintroduces implicit physics and generates incoherence.

The fundamental reorganization $\mathfrak{D}$, ordered by the structural index t and constrained by the limit c , maintains coherence without introducing physical time, causality, or propagation. The structural principle $E + C$ stabilizes configurations without resorting to energy, equations of motion, or physical

dynamics. From this reorganization emerge the continuous distance $D_S^{(e)}$ and the emergent correlation C_e , both non-observable but already regular.

The Φ maps are not ontological objects: they are compression relations that transform non-physical emergent constraints into coherent physical domains. From them, geometry, dynamics, fields, discrete quantities, statistical distributions, and stable configurations emerge simultaneously:

$$\Phi_\alpha: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow \mathcal{D}_{\text{physical}}^{(\alpha)}.$$

This multi-emergence is a unique property of the PSE: all physical domains derive from the same non-physical source, guaranteeing internal compatibility among geometry, dynamics, fields, discrete quantities, and statistics.

The structural simplicity of the PSE is a logical consequence: it is the simplest possible structure that does not introduce physical concepts at the ontological level. Any simpler theory is incoherent; any more complex theory is arbitrary. The PSE does not arise as an alternative: it arises as **what remains** once all forms of incoherence are eliminated.

In summary, the Theoretical Positioning shows that:

- the ontological chain is inevitable;
- multi-emergence is guaranteed;
- the structure is closed, minimal, and non-retroactive;
- no physical quantity is fundamental;
- observable physics is a compressed representation of emergent correlation;
- the PSE is scientifically assessable because it eliminates all forms of circularity and physical contamination.

The PSE is not a speculative proposal: it is the minimal form of ontological coherence required to describe physical emergence from a non-physical source.

Part IV - Verifiability

The verifiability of the PSE is grounded in the emergent nature of the physical domain. The structure $(D_S^{(e)}, C_e)$ produces metrics, fields, and dynamics as compressive representations through Φ . These representations possess observable forms that can be compared with experiments and empirical data.

The verifiability of the PSE is articulated along two dimensions:

- **emergent signatures**, which define specific observables;
- **anomalous data**, already detected, consistent with the emergent structure.

1 Emergent signatures and verifiable experiments

Emergent signatures are structural properties of the emergent domain that Φ expresses in the physical domain. They belong to three categories: geometric, correlative, and dynamic.

1.1 Geometric signature — Emergent metric saturation

The emergent distance is continuous, regular, and saturating. The representation Φ expresses this saturation as an upper bound on metric variations.

Emergent form High-frequency gravitational waves exhibit metric compression relative to the predictions of general relativity.

Observable Systematic deviation in the waveform at frequencies above the kilohertz regime.

Relevant experiments LIGO, Virgo, KAGRA, Einstein Telescope, Cosmic Explorer.

1.2 Quantum signature — Minimal granularity of emergent correlation

The emergent correlation possesses a minimal variation $\min \Delta C_e$. In the physical domain this variation appears as an action threshold.

Emergent form Ultra-low-energy quantum systems exhibit residual nonlinearity in superposition.

Observable Microscopic deviations from linearity in ultra-sensitive atomic interferometers.

Relevant experiments Atomic interferometry, Bose–Einstein condensates, MAQRO.

1.3 Correlative signature — Regular emergent nonlocality

The emergent correlation is nonlocal, continuous, and monotonic. The representation Φ expresses this form as regular nonlocal correlation.

Emergent form The decay of nonlocal correlation follows a monotonic regular form, distinct from that predicted by standard quantum mechanics.

Observable Systematic deviations in Bell tests at distances exceeding planetary scale.

Relevant experiments Micius satellite, terrestrial quantum networks, global-scale entanglement tests.

1.4 Dynamic signature — Emergent stability of fields

Emergent fields are saturating and free from uncontrolled divergences. Compressivity eliminates ultraviolet and infrared instabilities.

Emergent form In extreme regimes, fields exhibit stability superior to the predictions of field theory.

Observable Deviations in energy distributions in ultra-high-energy collisions or ultra-hot plasmas.

Relevant experiments

LHC, RHIC, Future Circular Collider.

1.5 Unity of emergent signatures

Geometric, correlative, and dynamic signatures are not independent: they represent different aspects of the same emergent structure $(D_S^{(e)}, C_e)$. Their coherence is a consequence of the pregeometric form and of the compressive nature of the Φ representation.

2 Anomalous observational data consistent with the PSE

This section does not attribute the explanation of observational phenomena to the PSE: it shows that **the emergent form of the PSE is compatible** with anomalies already detected across different areas of physics.

2.1 Cosmic acceleration

Anomaly The accelerated expansion of the universe is not explained by matter or radiation and requires the introduction of dark energy.

Consistency with the PSE The saturation of emergent distance produces a **natural geometric expansion**, represented by Φ as coherent metric dilation.

Emergent form The expansion is not caused by a physical source: it is a property of the saturating emergent structure.

Observables consistent with the PSE

- regular cosmic acceleration,
- absence of additional energetic sources,
- metrically coherent behaviour on cosmological scales.

Relevant experiments and observations

- Type Ia supernovae
- BAO (Baryon Acoustic Oscillations)
- Planck, WMAP
- Euclid, LSST

2.2 Hubble tensions

Anomaly Local and cosmological measurements of the Hubble constant do not coincide, producing a stable and unresolved tension.

Consistency with the PSE The compressivity of the Φ representation can generate **slightly different metric scales** in domains with different emergent correlation.

Emergent form The emergent structure is not perfectly homogeneous: regions with different correlation produce metric representations with slightly varying scales.

Observables consistent with the PSE

- systematic differences between local and global measurements,
- metric variations coherent with non-uniform emergent domains.

Relevant experiments and observations

- SH0ES
- Planck
- HST, JWST
- DESI

2.3 Anomalous quantum oscillations

Anomaly Ultra-sensitive quantum systems exhibit microscopic deviations from linear superposition.

Consistency with the PSE The minimal granularity $\min \Delta C_e$ produces **residual nonlinearities** at ultra-low energy.

Emergent form The emergent correlation possesses a minimal threshold that Φ represents as discrete variation of action.

Observables consistent with the PSE

- oscillations not perfectly linear,
- regular deviations in atomic interferometers,
- threshold effects in low-energy systems.

Relevant experiments and observations

- atomic interferometry (Stanford, MIT, LENS)
- Bose–Einstein condensates
- macroscopic superposition tests (MAQRO)

2.4 Anomalous stability of high-energy plasmas

Anomaly Ultra-hot plasmas exhibit stability superior to field-theoretic predictions.

Consistency with the PSE Saturation and compressivity eliminate UV divergences, producing **more stable emergent fields**.

Emergent form The Φ representation compresses extreme variations, generating regular fields even in high-energy regimes.

Observables consistent with the PSE

- plasma stability beyond QFT predictions,
- regular energy distributions,
- absence of uncontrolled divergences.

Relevant experiments and observations

- LHC Run 3
- RHIC
- Future Circular Collider

2.5 Quantum nonlocality on planetary scale

Anomaly Quantum correlations over distances > 1000 km exhibit extremely regular and stable behaviour.

Consistency with the PSE The emergent correlation is **nonlocal and monotonic**, with regular form over large distances.

Emergent form Nonlocality is not a pointwise phenomenon: it is a global property of the emergent correlation.

Observables consistent with the PSE

- regular nonlocal correlations,
- stability of correlation over planetary distances,
- coherent monotonic decay.

Relevant experiments and observations

- Micius satellite
- terrestrial quantum networks
- global-scale Bell tests

2.6 Anomalous distribution of dark matter

Anomaly Galactic rotation curves, gravitational lensing, and mass distributions do not match visible matter. No experimental evidence for dark-matter particles exists despite decades of searches.

Consistency with the PSE The emergent structure $(D_S^{(e)}, C_e)$ contains regions with **non-uniform structural rigidity**. The Φ representation expresses this rigidity as:

- coherent metric variations,
- gravitational effects without additional mass,
- emergent distributions that mimic dark-matter halos.

The emergent correlation produces **shape gradients** that appear in the physical domain as invisible mass distributions.

Emergent form Dark matter is not a particle or an additional field: it is an **elastic property of the emergent structure**, represented by Φ as a geometric contribution to the physical metric.

Observables consistent with the PSE

- regular galactic rotation curves,
- gravitational lensing without luminous sources,
- non-Gaussian mass distributions,

- halos with cored rather than cuspy profiles.

Relevant experiments and observations

- Gaia
- JWST
- DES, LSST, Euclid
- Bullet Cluster and analogous systems

2.7 Baryon Asymmetry

Anomaly The universe contains significantly more matter than antimatter, despite physical laws being largely symmetric under matter–antimatter exchange. No experimental evidence exists for mechanisms capable of producing the observed imbalance.

Consistency with the PSE The emergent structure $(D_S^{(e)}, C_e)$ can produce **asymmetric correlation domains**, where the representation Φ expresses correlation gradients as effective matter–antimatter imbalance.

Emergent form Baryon asymmetry is not a fundamental violation: it is a **structural asymmetry** of the emergent correlation.

Observables consistent with the PSE

- absence of large-scale antimatter domains,
- uniform matter dominance across cosmological scales,
- correlation patterns consistent with asymmetric emergent domains.

Relevant experiments and observations

- CMB anisotropies (Planck)
- BBN (Big Bang Nucleosynthesis)
- LHCb (CP violation studies)
- Belle II

2.8 Macroscopic quantum coherence

Anomaly Superconductors, superfluids, and Bose–Einstein condensates exhibit coherent quantum behaviour over macroscopic distances, far beyond typical quantum-mechanical correlation lengths.

Consistency with the PSE The emergent correlation C_e is **continuous and monotonic**, allowing coherent behaviour over extended domains. Φ represents this coherence as macroscopic quantum order.

Emergent form Macroscopic coherence is a **natural property of the emergent correlation**, not an exceptional quantum phenomenon.

Observables consistent with the PSE

- long-range phase coherence,

- stable macroscopic superposition,
- regular correlation decay in extended quantum systems.

Relevant experiments and observations

- superconductivity and superfluidity
- BEC coherence measurements
- SQUID interferometry
- macroscopic entanglement experiments

2.9 Large-scale cosmic regularity (> 1 Gpc)

Anomaly The distribution of matter on gigaparsec scales is more regular and coherent than predicted by Λ CDM, with structures such as filaments and walls showing unexpected uniformity.

Consistency with the PSE The emergent correlation is **monotonic and continuous**, producing large-scale regularity in the representation Φ .

Emergent form Cosmic regularity is a **geometric expression of the emergent correlation**, not a statistical anomaly.

Observables consistent with the PSE

- coherent filaments and walls,
- regularity of large-scale structures,
- deviations from purely stochastic clustering.

Relevant experiments and observations

- SDSS
- DESI
- Euclid
- LSST

2.10 Anomalies in strong gravitational lensing

Anomaly Several strong-lensing systems show deviations from GR+ Λ CDM predictions, including anomalous magnification ratios and unexpected mass distributions.

Consistency with the PSE Regions of **non-uniform structural rigidity** in the emergent domain produce metric variations that Φ represents as lensing anomalies.

Emergent form Strong-lensing deviations are **elastic effects of the emergent structure**, not evidence of exotic matter.

Observables consistent with the PSE

- anomalous flux ratios,
- lensing without sufficient visible mass,
- coherent distortions in cluster lenses.

Relevant experiments and observations

- JWST
- HST strong-lensing surveys
- Euclid lensing catalogue
- SLACS

2.11 Low-energy neutrino oscillation anomalies

Anomaly Some low-energy neutrino experiments show oscillation patterns that deviate slightly from standard three-flavour predictions.

Consistency with the PSE The minimal granularity $\min \Delta C_e$ introduces **threshold effects** that Φ represents as small deviations in oscillation behaviour.

Emergent form Neutrino oscillations reflect **granular structure in the emergent correlation**, not new neutrino species.

Observables consistent with the PSE

- non-linear oscillation phases at low energy,
- threshold-like behaviour,
- regular deviations from standard mixing angles.

Relevant experiments and observations

- Borexino
- SNO+
- IceCube low-energy data
- JUNO (future)

2.12 Uniformity of the Cosmic Microwave Background (CMB)

Anomaly The CMB exhibits an extremely high level of uniformity between regions that, according to standard cosmological geometry, could not have been in causal contact before the emission of the radiation. This homogeneity requires the introduction of cosmic inflation as an additional dynamical phase.

Consistency with the PSE In the PSE, the emergent structure $(D_S^{(e)}, C_e)$ contains global pregeometric correlations. The compressive nature of the representation Φ produces metric domains that are already coherent on large scales, without requiring an inflationary dynamical mechanism. The uniformity of the CMB is therefore compatible with the presence of global emergent correlations preceding physical geometry.

Emergent form The homogeneity does not arise from a physical process of accelerated expansion, but from the global nature of the emergent correlations that Φ compresses into a uniform metric.

Observables consistent with the PSE

- large-scale isotropy of the CMB,

- temperature coherence between regions not causally connected,
- stability of primordial anisotropies,
- no need for an inflaton field or a dedicated potential.

Relevant experiments and observations

- COBE
- WMAP
- Planck
- future CMB-S4 missions

2.13 Absence of observational data contradicting the PSE

To date, no observational data are known that require a physical structure incompatible with the emergent form $(D_S^{(e)}, C_e)$ or with the compressive nature of the representation Φ . This absence of negative data does not imply that the PSE is correct, but it indicates that the emergent framework remains empirically compatible with the available measurements.

Part V - Ontological Vision and Perspectives

The Structural Principle of Entanglement (PSE) presents a complete and self-sufficient ontological framework in which fundamental reality is described by a discrete prespatial structure, devoid of geometry, causality, and physical quantities. From this structure emerge the continuous structural distance, the emergent correlation, and (through the monotonic and non-invertible projection Φ) all observable physical quantities.

The theory is coherent, non-retroactive, and non-invertible: each level contains less information than the previous one, and no physical quantity can reconstruct the fundamental structure. Physics is not assumed: it is a compressed representation of emergent correlation, not an ontological property of reality.

1. Closure of the Conceptual Framework

The conceptual framework of the PSE is complete. The following have been defined:

Ontological Elements

- the prespatial structure S ;
- the fundamental structural distance $D_S^{(f)}$;
- the fundamental correlation C_f ;
- the emergent distance $D_S^{(e)}$;
- the emergent correlation C_e .

Fundamental Ontological Constraints

- the structural index t (discrete ordering of reorganization);
- the structural principle $E + C$ (stabilization of configurations);
- the threshold Δ (selection of fundamental correlations);
- the structural limit c (local coherence of reorganization).

Necessary Operators (Non-Ontological)

- the aggregation and regularization operators F_λ and G_λ ;
- the compressive projection Φ , which formalizes $(X, D_S^{(e)}, C_e)_{\text{reg}}$ into the physical domain.

Non-Ontological Result

- the observable physical domain $\mathcal{D}_{\text{physical}}$.

Epistemic Aspects

- the general properties of the physical domain;
- indirect verifiability.

2. Role of Mathematical Formalization

The mathematical formalization of the PSE is the natural continuation of the conceptual framework. It allows one to:

- rigorously define the fundamental discrete structure;
- characterize the threshold Δ and the selectivity of correlations;
- construct the emergent structural distance $D_S^{(e)}$ through F_λ ;
- construct the emergent correlation C_e through G_λ ;
- define the monotonic (non-physical) function F ;
- formalize the properties of the compressive projection Φ ;
- prove theorems on the ontological hierarchy and on non-invertibility.

Mathematics does not introduce new concepts: it implements those already defined in the ontological framework.

3. General Ontological Vision

The PSE assumes as its ontological basis the prespatial structure S , composed of pure relational entities, devoid of any physical concept, connected by binary correlations and reorganized by the fundamental process $\mathfrak{D}$, stabilized by the structural principle $E + C$.

S is:

- a mathematical point without extension,
- the whole, because no “outside” exists,
- the nothing, because it contains no physical concept,
- the only possible ontological domain.

The increase in complexity of S generates a sequence of emergent levels: pure correlations, emergent coherence, emergent structure, collective rigidity, geometric compression, physical metric, physical dynamics, matter, chemistry, biology, consciousness.

Physics is not the foundation of existence: it is an emergent level of the complexity of S .

4. S as Ontological Source of Emergent Domains

S contains the ontological possibility of multiple emergences: stable, metastable, unstable, ephemeral, long-lasting, self-collapsing, and self-expanding domains.

Each emergent domain may possess:

- different constants,
- different geometries,
- different metrics,
- different dynamics,

- different symmetry breakings,
- possibility or impossibility of chemistry and biology.

These domains are not regions of space: they are emergent configurations of the complexity of S , not physical locations.

5. Physical and Conceptual Consequences

The PSE offers new interpretative keys for physical phenomena:

- entanglement as a trace of primordial nonlocality;
- indeterminacy as a property of geometric compression;
- wave–particle duality as geometric selection of emergent patterns;
- uniformity of the CMB as a consequence of the prespatial phase;
- dark matter as non-localized rigidity;
- dark energy as emergent geometric tension;
- consciousness as a non-local emergent pattern.

6. Future Directions

The main directions of development include:

- advanced metric extensions;
- emergent dynamical structures;
- emergent fields and interactions;
- correlational phenomenology;
- discrete simulations of the fundamental domain;
- experimental verifiability through strongly correlated quantum states;
- holographic and informational applications.

7. Epistemic Value of the PSE

The PSE represents:

- a unique ontological foundation,
- a natural emergent structure,
- a conceptual reconciliation between quantum and relativistic phenomena,
- an epistemic simplification of modern physics.

The PSE does not replace physics: it clarifies its place in the order of existence.

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